

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

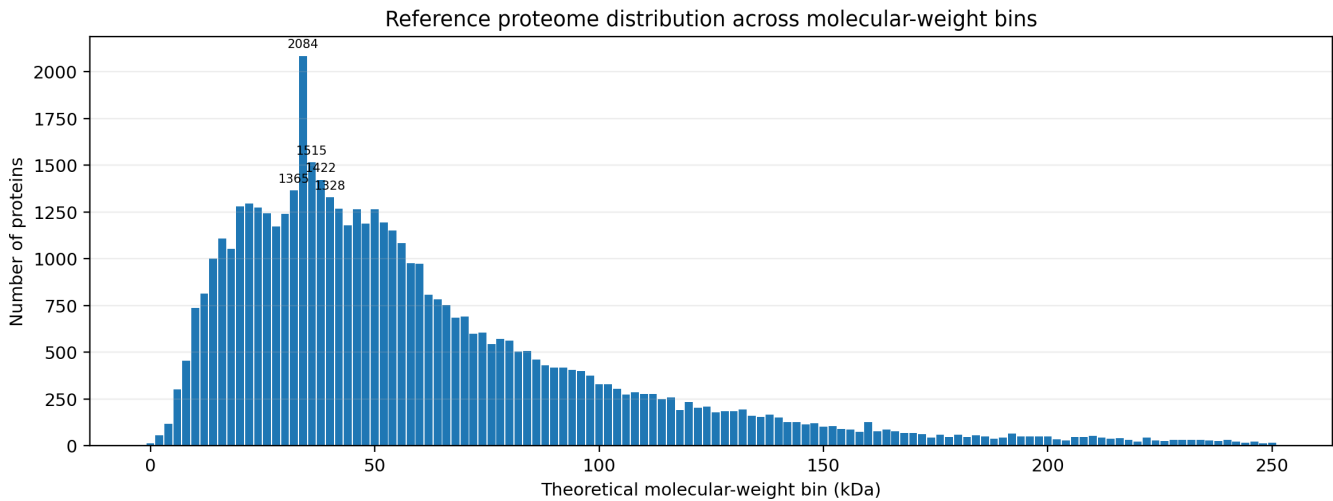
Metric	Value
Proteins parsed	51,612
Molecular-weight bins	304
Bin width	2 kDa
MW range	0.97 to 891.87 kDa
Median MW	50.05 kDa
Mean MW	65.81 kDa
Proteins <= 50 kDa	25,775 (49.9%)
Proteins 50-100 kDa	17,162 (33.3%)
Proteins > 100 kDa	8,675 (16.8%)
High-priority bins	31
Elevated-priority bins	45
Routine-priority bins	228
Source FASTA	cow_UP000009136.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

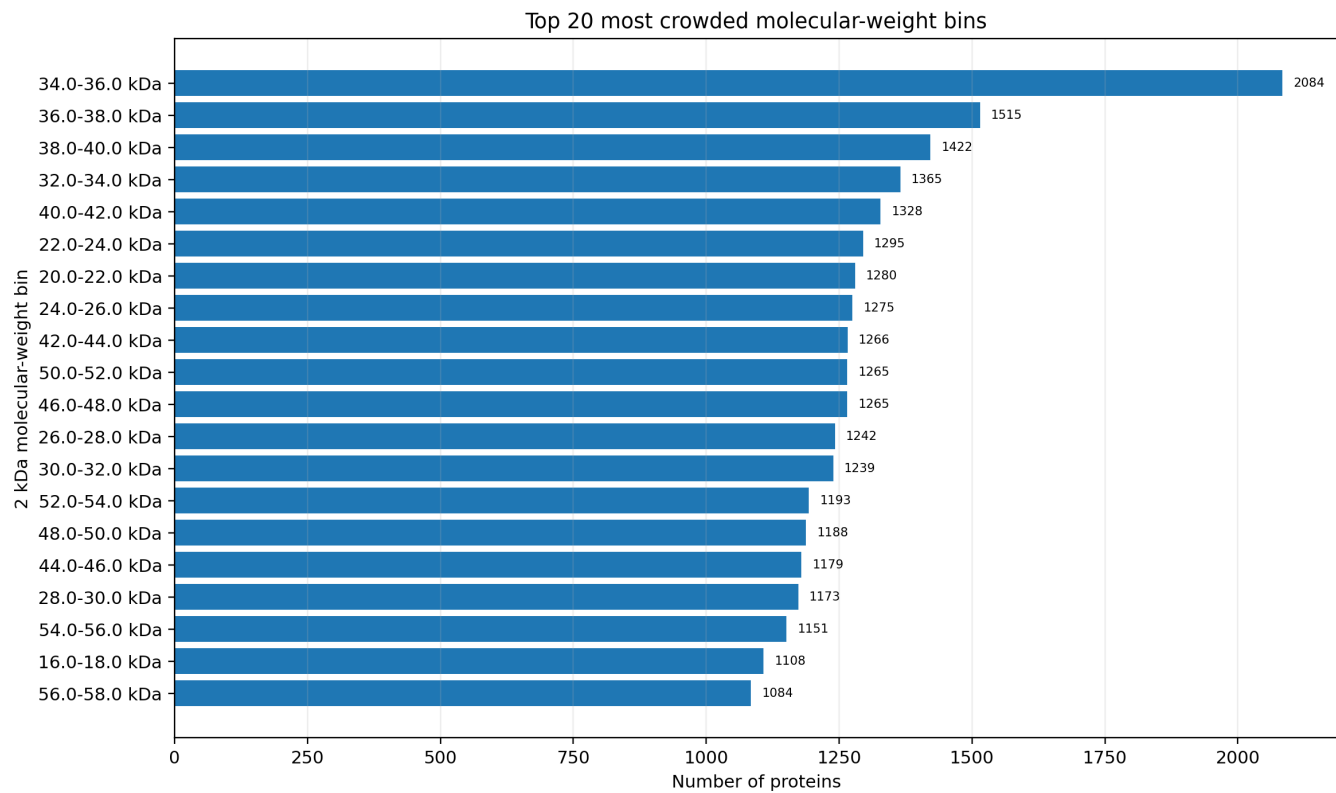
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

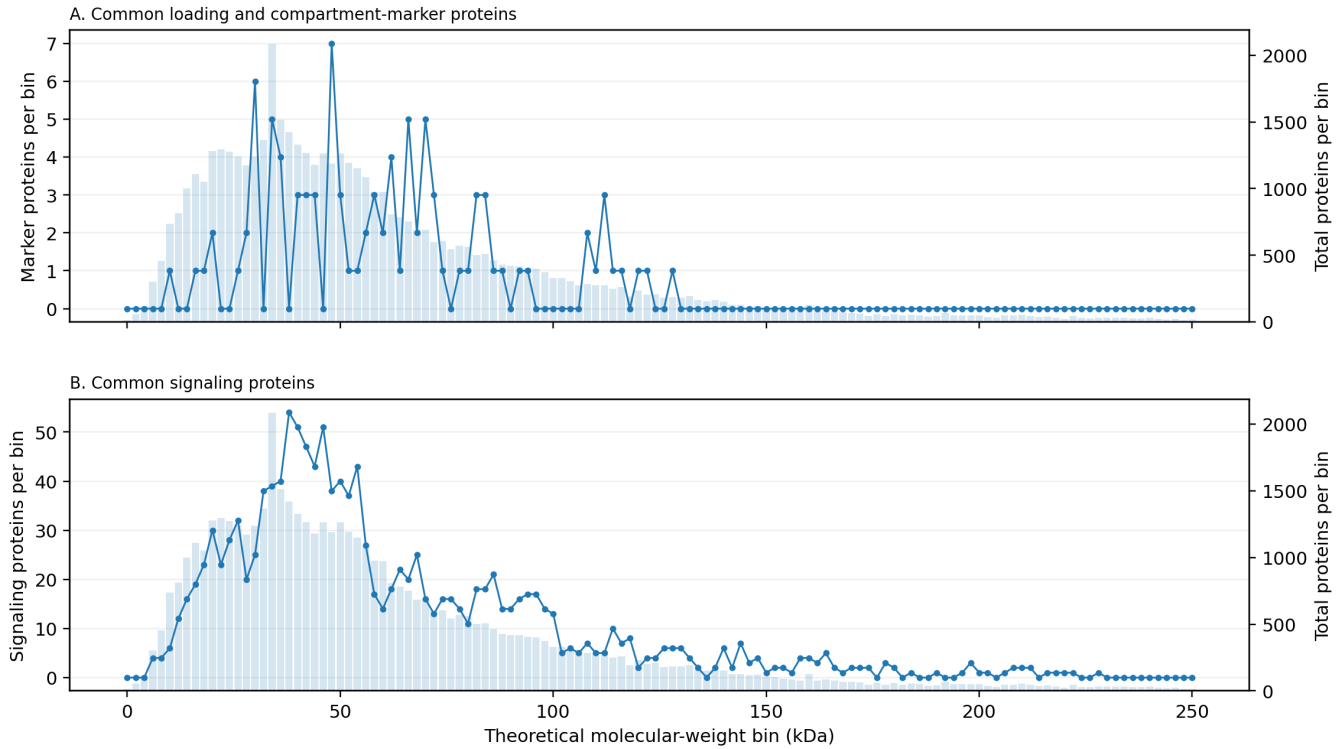
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

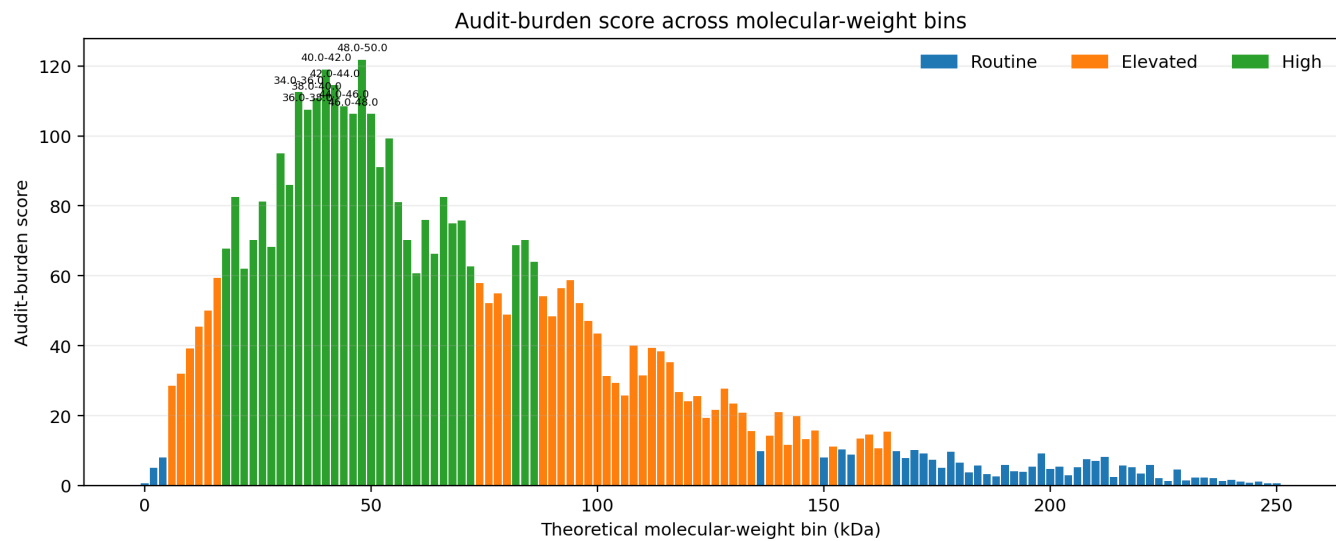
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
48.0-50.0 kDa	1188	7	38	high	121.8
40.0-42.0 kDa	1328	3	51	high	119.0
42.0-44.0 kDa	1266	3	47	high	114.5
34.0-36.0 kDa	2084	5	39	high	112.5
38.0-40.0 kDa	1422	0	54	high	110.8
44.0-46.0 kDa	1179	3	43	high	108.5
36.0-38.0 kDa	1515	4	40	high	107.5
46.0-48.0 kDa	1265	0	51	high	106.2
50.0-52.0 kDa	1265	3	40	high	106.2
54.0-56.0 kDa	1151	1	43	high	99.2
30.0-32.0 kDa	1239	6	25	high	95.0
52.0-54.0 kDa	1193	1	37	high	91.0
32.0-34.0 kDa	1365	0	38	high	86.0
20.0-22.0 kDa	1280	2	30	high	82.5
66.0-68.0 kDa	753	5	20	high	82.5
26.0-28.0 kDa	1242	1	32	high	81.2
56.0-58.0 kDa	1084	2	27	high	81.0
62.0-64.0 kDa	809	4	18	high	76.0
70.0-72.0 kDa	690	5	16	high	75.8
68.0-70.0 kDa	686	2	25	high	75.0
24.0-26.0 kDa	1275	0	28	high	70.2
58.0-60.0 kDa	976	3	17	high	70.2
84.0-86.0 kDa	508	3	18	high	70.2
82.0-84.0 kDa	503	3	18	high	68.8
28.0-30.0 kDa	1173	2	20	high	68.2

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
48.0-50.0 kDa	1188	7	38	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	CEACAM16, PICK1, ECMX, PSKH1, PRKN, PNLI, PRP3, IFRD2, MAEL, SYT11, TFAP2C, LOC112443216 [Tubulin family], TRMT11, CCT6B, NTSDC1, TERC2 [Tubulin family], STK25, SDC3, SERPINB12
40.0-42.0 kDa	1328	3	51	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	CD44, , PGA, IRF2, SLC35C1, ACKR4, ST3GAL2, ADORA2B, ZDHHC12, TIMM50 [Mitochondrial markers], GMPR2, ERI1, PPHL1N, PCBP4, HRH2, NELLE, RBM4B, UBXN11
42.0-44.0 kDa	1266	3	47	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	RPUSD4 [Mitochondrial markers], DTWD1, DHODH [Mitochondrial markers], ELF3, PEX26, , GARIN2, PTX3, ACTA2 [Actin family], ZAR1, LOC112443864, CEP57L1, METTL18, SLC35F1, TEX9, WDR31, TNIP3 [Actin family], ACTC1 [Actin family]
34.0-36.0 kDa	2084	5	39	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	LOC526989, NIT1, SYP, SGTA, MS4A1, IGFBP2, ZFP36L1, VTA1, OR5B32, MYG1, DHRS1, GPM6B, CCDC69, CDK1 [Cell-cycle CDK/cyclin], PIM3, TMX2, SEC13, TAL1
38.0-40.0 kDa	1422	0	54	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	NTM, LCMT1, EFNB1, REM2, DTNBP1, AMT, ELAVL2, SCAMP1, SLX1A, TMEM268, OTULIN, YOD1, RDH16, HAPLN3, PSM4, SGK2, GMPR2, PAQR5
44.0-46.0 kDa	1179	3	43	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	PHLDA1, NIPAL4, BDKR2, HNRNPDL, NCK2, KHDRBS1, HEMK1, VDR, DEPTOR [Actin family], CASP1 [Apoptosis caspase/BCL], BSCL2, ERLEC1, PRR5L, PELI2, MAPK8 [MAPK cascade], , WTAF, KIN
36.0-38.0 kDa	1515	4	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	high	PLCXD1, HLF, TMEM231, MSRB3, HNRNPA2B1, NDEL1, HNRNPAB, HNRNPA2B1, PHYH, LOC132342121, OR6C7G, C1QTNF7, STBD1, ASB7, FOXR2, AUH, C19H17orf58, LDHC
46.0-48.0 kDa	1265	0	51	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	DAPK2, , CACNB4, LOC525484, POGLUT1, BABAM2, PAAF1, BEND7, GPR161, ASB18, SERPINA7, WDR62, MKRN2, RPL3, RUFY2, NCEH1, GNA13, METAP1
50.0-52.0 kDa	1265	3	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TMEM184C, MEIKIN, CDSN, RCC1L, C20H5orf22, RGS7, ZNF362, OR6Z9, CHGA, TMPRSS2, SOX10, SLC2A8, CAMKV, EXOC3L2, TUBD1 [Tubulin family], KRT7, SERGEF, NDE1
54.0-56.0 kDa	1151	1	43	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	WDR37, POGLUT2, GLRB, BCAS1, NYX, ZNF19, CDH8, LTA4H, SCARB2, SPRED2, TTBK2, ANXA11, SLC37A3, CASP16 [Apoptosis caspase/BCL], SLC2A3, CCDC85A, JAKMIP1 [Actin family; JAK-STAT], SLC38A1
30.0-32.0 kDa	1239	6	25	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	ANXA8L1, SHFL, ARF3, TNFRSF13B, , PSMB7, THAP1, , TATDN3, SPATS1, LENG1, PLP1, POLR3F, PRSS33, RPL7L1, MS4A10, ATPSCKMT, RPL7A
52.0-54.0 kDa	1193	1	37	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TIMM44 [Mitochondrial markers], PPP2R2B, EOGT, IFFO2, FKTN, GLRA2, ATG4A, DDX6, RMND1, LOC786377, PPP2R2B, ACP1, IRF6, ENTREP2, LETM2, ERG, MEOB, FADS3
32.0-34.0 kDa	1365	0	38	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	MRRF [Cell-cycle CDK/cyclin; Mitochondrial markers], BOLA-DQB, ZNF691, AGPAT1, NKX2-6, PGAM5 [Mitochondrial markers], KDELR3, CD47, DTWD2, SOD2 [Mitochondrial markers], MESD, COPE, SLA2, PDCD1LG2, SAMD10, SAP30BP, SLC25A21 [Mitochondrial markers], TRIM52
20.0-22.0 kDa	1280	2	30	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LOC100847317, C19H17orf49, SLC25A26 [Mitochondrial markers], CDKN2AIPNL [Actin family; Cell-cycle CDK/cyclin], LOC514011, GH1, IL1RN, CSN3, MORNS, EPO, DUT, AP1S3, JDP2, IL7, MECP2, PFDN1, LOC104970886, CATHL2
66.0-68.0 kDa	753	5	20	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	, ZNF181, SCFD1, PRKCI, CENPB, PATL2, RTKN, ASMTL, HMGXB4, CCDC38, GRSF1, MELK, JPH4, PTPN11, SAMHD1, GPR153, RAD17, CBF2A23
26.0-28.0 kDa	1242	1	32	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UBTD1, DIABLO [Mitochondrial markers], C8H9orf152, EMC10, LDHC, CCDC124, TSPAN13, MPV17L2 [Mitochondrial markers], FAM76A, MRPS18A, ELSPBP1, KCTD11, SCN1B, CSN1S2, MT-CO2 [Mitochondrial markers], , TM4SF1, RND1
56.0-58.0 kDa	1084	2	27	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	RBP1, CHRNA7, KCNF1, FSD1, VEZF1, GTF2H2, SLC1A4, XYLB, AAAS, ZNF775, USP14, MBOAT1, FTO, STK3, TUBB3 [Tubulin family], MRP59, CCDC50, UBE3D
62.0-64.0 kDa	809	4	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	ALDH4A1, MTUS1, TDRKH, DCAF15, LOC786974, CFAP53 [p53 pathway], RINGT1, LMBRD1 [Lamin family], CYP2U1, SPPL2A, XRR1A, EPHX2, ZUP1, FBXW8, SLC1A2, PLB01, LRRC47, EPN2
70.0-72.0 kDa	690	5	16	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers	high	ZWILCH, ZDHHC8, OLFML2A, ENG, ATG16L1, EYA4, MTM1, SEMA7A, FUBP1, SASS6, RLIM, KLHL31, FNBP1L, MED26, SLC6A4, TKTL2, ACOT11, DMWD
68.0-70.0 kDa	686	2	25	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	PLK1, BCAM, TCF7L2, AKAP8L, EPB42, BTRC, MCOLN2, ZNF792, SSX2IP [Actin family], DET1, CFI, LMBRD1 [Lamin family], C10H14orf39, LOC112442728, GNL3L, SLC6A6, KNG1, LRRC27
24.0-26.0 kDa	1275	0	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	INSL6, APOO, NME5, CRIP3, , LOC781977, NDUBF9, PDGFA, NAT9 [Tubulin family], DPT, CD9, LHFPL3, TATDN1, FAAP24, TMEM114, ADAMTSL1, VT1A, C15H11orf52
58.0-60.0 kDa	976	3	17	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	NEPN, LOC786974, GHDC, HHATL, LOC522334, TRAF3, ANGPTL1, TUBE1 [Tubulin family], KLF15, TMEM175, CYB5R4, SLC16A3, SYT7, LOC132344484, ETV3, KCNA10, GLCC11, MEPE
84.0-86.0 kDa	508	3	18	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	PIK3R6, SNAP91, MID2, IQSEC3, ADD1, FCHSD2, KIAA0319L, AMFR, HSD17B4, LOC515755, ARHGAP10, CAPN11, NCSTN, CUL4B, LOXL4, CNMNA, SYTL5, DRC1
82.0-84.0 kDa	503	3	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers	high	RPS6KA1, LPO, DENND2B, RNF214, TRAF4, ZC3H14, DNMT3B, DNMT3B, ARNT, FGFR1, LRSAM1, PUM3, SP3, CPXM2, CDH7, MAP7, XRR1A, RAP1GAP2
28.0-30.0 kDa	1173	2	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	LRRC3, SINHCAF, PRP4 [Actin family], SPINT2, CMBL, LOC505326, , LRRC61, PDCL2, IL1A, FAM3B, PACRGL, YIPF5, SCAMP5, RWDD1, UCHL1, STXB6, PMP6A1
18.0-20.0 kDa	1052	1	23	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RAB7B, NCBP2, ABCC9, MFAP5, UBE2I, CIRBP, RFESD, SAMD5, LRCOL1, ZBTB80S, DCTN3 [Actin family], YY1, ALOX5AP, IER3, RAB18, SREK1IP1, TAF2A, NAMPT
64.0-66.0 kDa	784	1	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	ARHGEF3, SLC22A16, FAM151A, PRKAA1, WBP11, FAM151A, NF2, TGFBR2, LOC785216, ARIH1, PINK1, NEURL1, LMF1, LRRIQ4, RBM47, NT5C1B, IL27RA,
86.0-88.0 kDa	461	1	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	HSPA14 [Heat shock proteins], ELAC2, VPSS1, ZNF667, CHPF2, POT1, SMURF1, LOC112445144, ARHGAP10, ZNF184, GFM2 [Mitochondrial markers], ACAP1, LOC515755, EZH1 [Histone family], LOC518612, IDUA, CUX1, PCSK7
72.0-74.0 kDa	598	3	13	Actin family; Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	IRAK2, LRCH4, FBXL18, XRC6B, PTH1R, EPB41L4A, SOX13, SLC6A14, SH2B1, DDX43, LOC112445770, PLPPRA, HNRNPM, PRKCG, KIFBP, FUBP1, ODAD3, RAB11FIP4 [Actin family]
22.0-24.0 kDa	1295	0	23	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	BMERB1, COQ10B [Mitochondrial markers], SAMD5, LOC100298530, ARFIP1 [Actin family], RHOC, LOC132344423, CLDN11, SAT2, PEXT [Lamin family], FHL1, PPIF [Mitochondrial markers], TRIR [Actin family], CHMP1B, C17H5orf52, RAB1B, RNASE11, MGMT

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
11.21	RPLP0	A0AAA9T0W1	A0AAA9T0W1_BOVIN	Large ribosomal subunit protein uL10	10.0-12.0 kDa
16.30	TOMM20	A6H7B1	TOM20_BOVIN	Mitochondrial import receptor subunit TOM20 homolog	16.0-18.0 kDa
19.57	COX4I1	P00423	COX41_BOVIN	Cytochrome c oxidase subunit 4 isoform 1, mitochondrial	18.0-20.0 kDa
20.24	H2AC1	A0AAA9THX7	A0AAA9THX7_BOVIN	Histone H2A	20.0-22.0 kDa
20.62	COX4I1	A0AAF6Z2H9	A0AAF6Z2H9_BOVIN	Cytochrome c oxidase subunit 4	20.0-22.0 kDa
27.56	RPLP0	A0AAA9TK96	A0AAA9TK96_BOVIN	Large ribosomal subunit protein uL10	26.0-28.0 kDa
28.61	VDAC2	A0AAF6YY06	A0AAF6YY06_BOVIN	Non-selective voltage-gated ion channel VDAC2	28.0-30.0 kDa
28.75	PCNA	Q3ZBW4	PCNA_BOVIN	DNA sliding clamp PCNA	28.0-30.0 kDa
30.62	VDAC1	A0A3Q1MJQ7	A0A3Q1MJQ7_BOVIN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
30.74	VDAC1	P45879	VDAC1_BOVIN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
31.56	VDAC2	A0AAA9TU27	A0AAA9TU27_BOVIN	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
31.62	VDAC1	A0AAF6ZJG9	A0AAF6ZJG9_BOVIN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
31.62	VDAC2	P68002	VDAC2_BOVIN	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
32.00	VDAC1	A0A3Q1MS21	A0A3Q1MS21_BOVIN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
34.37	RPLP0	Q95140	RLA0_BOVIN	Large ribosomal subunit protein uL10	34.0-36.0 kDa
34.40	RPLP0	F1MDN4	F1MDN4_BOVIN	60S acidic ribosomal protein P0	34.0-36.0 kDa
35.22	TBP	Q2HJ52	TBP_BOVIN	TATA-box-binding protein	34.0-36.0 kDa
35.58	GAPDH	A0A3Q1M8J9	A0A3Q1M8J9_BOVIN	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
35.87	GAPDH	P10096	G3P_BOVIN	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
36.11	TBP	A0AAA9RYY7	A0AAA9RYY7_BOVIN	TATA-box binding protein	36.0-38.0 kDa
36.63	TUBB2A	G3X7R7	G3X7R7_BOVIN	Tubulin beta chain	36.0-38.0 kDa
36.67	TUBB2B	G3N1W7	G3N1W7_BOVIN	Tubulin beta chain	36.0-38.0 kDa
37.03	TBP	A0AAF6ZDS6	A0AAF6ZDS6_BOVIN	TATA-box binding protein	36.0-38.0 kDa
40.50	CALR	A0AAA9SSN4	A0AAA9SSN4_BOVIN	Calreticulin	40.0-42.0 kDa
41.74	ACTB	P60712	ACTB_BOVIN	Actin, cytoplasmic 1	40.0-42.0 kDa
41.94	TUBB2B	A0A452DI48	A0A452DI48_BOVIN	Tubulin beta chain	40.0-42.0 kDa
42.09	ACTB	A0AAA9SMJ2	A0AAA9SMJ2_BOVIN	Actin beta	42.0-44.0 kDa
43.12	ACTB	A0AAA9RYR7	A0AAA9RYR7_BOVIN	Actin beta	42.0-44.0 kDa
43.45	CALR	A0AAA9SYV6	A0AAA9SYV6_BOVIN	Calreticulin	42.0-44.0 kDa
44.34	ACTB	A0ACH6R3D3	A0ACH6R3D3_BOVIN	Actin beta	44.0-46.0 kDa
45.08	CALR	A0AAA9S1K1	A0AAA9S1K1_BOVIN	Calreticulin	44.0-46.0 kDa
45.78	TUBB2A	E1B953	E1B953_BOVIN	Tubulin beta chain	44.0-46.0 kDa
48.04	CALR	P52193	CALR_BOVIN	Calreticulin	48.0-50.0 kDa
48.10	CALR	F6QWE4	F6QWE4_BOVIN	Calreticulin	48.0-50.0 kDa
48.79	CANX	A0AAA9S391	A0AAA9S391_BOVIN	Calnexin	48.0-50.0 kDa
49.86	TUBA1C	Q3ZCJ7	TBA1C_BOVIN	Tubulin alpha-1C chain	48.0-50.0 kDa
49.91	TUBB2A	E1BJB1	E1BJB1_BOVIN	Tubulin beta chain	48.0-50.0 kDa
49.95	TUBB2B	Q6B856	TBB2B_BOVIN	Tubulin beta-2B chain	48.0-50.0 kDa
49.97	TUBB2B	A0ABI0P3K8	A0ABI0P3K8_BOVIN	Tubulin beta chain	48.0-50.0 kDa
50.14	TUBA1A	F1MNF8	F1MNF8_BOVIN	Tubulin alpha chain	50.0-52.0 kDa
50.43	TUBB3	Q2T9S0	TBB3_BOVIN	Tubulin beta-3 chain	50.0-52.0 kDa
50.46	TUBA1B	A0A452DIK7	A0A452DIK7_BOVIN	Tubulin alpha chain	50.0-52.0 kDa
53.86	HSPA8	A0AAA9RT91	A0AAA9RT91_BOVIN	Heat shock protein family A (Hsp70) member 8	52.0-54.0 kDa
55.68	LMNA	A0AAA9TES4	A0AAA9TES4_BOVIN	Prelamin-A/C	54.0-56.0 kDa
56.03	TUBB3	A0AAF6YHE7	A0AAF6YHE7_BOVIN	Tubulin beta chain	56.0-58.0 kDa
57.27	P4HB	P05307	PDIA1_BOVIN	Protein disulfide-isomerase	56.0-58.0 kDa
59.58	CANX	A0AAA9TII8	A0AAA9TII8_BOVIN	Calnexin	58.0-60.0 kDa
59.69	ATP5F1A	F1MLB8	F1MLB8_BOVIN	ATP synthase subunit alpha	58.0-60.0 kDa
59.72	ATP5F1A	P19483	ATPA_BOVIN	ATP synthase F(1) complex subunit alpha, mitochondrial	58.0-60.0 kDa
60.25	P4HB	A0AAA9SHU1	A0AAA9SHU1_BOVIN	Protein disulfide-isomerase	60.0-62.0 kDa
60.61	CANX	A0AAA9SMV6	A0AAA9SMV6_BOVIN	Calnexin	60.0-62.0 kDa
62.64	CANX	A0AAA9TOX0	A0AAA9TOX0_BOVIN	Calnexin	62.0-64.0 kDa
63.35	P4HB	F6Q9Q9	F6Q9Q9_BOVIN	Protein disulfide-isomerase	62.0-64.0 kDa
63.43	CANX	A0AAA9TTD5	A0AAA9TTD5_BOVIN	Calnexin	62.0-64.0 kDa
63.80	SDHA	A0AAA9TB48	A0AAA9TB48_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	62.0-64.0 kDa
64.57	CANX	A0AAA9TTL3	A0AAA9TTL3_BOVIN	Calnexin	64.0-66.0 kDa
66.34	LMNB1	A0A3Q1LNG7	A0A3Q1LNG7_BOVIN	Lamin B1	66.0-68.0 kDa
66.42	LMNB1	A0A3Q1MFL8	A0A3Q1MFL8_BOVIN	Lamin B1	66.0-68.0 kDa
66.77	HSPA8	A0A3Q1M8L6	A0A3Q1M8L6_BOVIN	Heat shock cognate 71 kDa protein	66.0-68.0 kDa
67.53	SDHA	A0A3Q1MCH3	A0A3Q1MCH3_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	66.0-68.0 kDa
67.78	CANX	G3MY89	G3MY89_BOVIN	Calnexin	66.0-68.0 kDa
69.02	HSPA1A	A0AAF6YYN3	A0AAF6YYN3_BOVIN	Heat shock protein family A (Hsp70) member 1A	68.0-70.0 kDa
69.08	LMNB1	A0AAA9TNY2	A0AAA9TNY2_BOVIN	Lamin-B1	68.0-70.0 kDa
70.26	HSPA1A	Q27975	HS71A_BOVIN	Heat shock 70 kDa protein 1A	70.0-72.0 kDa
70.65	LMNA	A0AAA9SP53	A0AAA9SP53_BOVIN	Prelamin-A/C	70.0-72.0 kDa
70.72	HSPA8	A0AAA9TJ30	A0AAA9TJ30_BOVIN	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
71.24	HSPA8	P19120	HSP7C_BOVIN	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
71.76	HSPA8	A0AAF6YYH1	A0AAF6YYH1_BOVIN	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.26	HSPA8	A0A3Q1LMS5	A0A3Q1LMS5_BOVIN	Heat shock cognate 71 kDa protein	72.0-74.0 kDa
72.94	SDHA	P31039	SDHA_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
73.06	SDHA	G3MY67	G3MY67_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
74.13	LMNA	F1MYG5	F1MYG5_BOVIN	Prelamin-A/C	74.0-76.0 kDa
79.18	LMNA	A0AAA9SIS5	A0AAA9SIS5_BOVIN	Prelamin-A/C	78.0-80.0 kDa
81.72	HSP90AB1	G5E507	G5E507_BOVIN	Heat shock protein 90 alpha family class B member 1	80.0-82.0 kDa
82.31	HSP90AB1	A0AAA9TAX9	A0AAA9TAX9_BOVIN	Heat shock protein 90 alpha family class B member 1	82.0-84.0 kDa

MW kDa	Gene	Accession	Entry	Protein	MW bin
83.04	HSP90AB1	A0AAA9SQM8	A0AAA9SQM8_BOVIN	Heat shock protein 90 alpha family class B member 1	82.0-84.0 kDa
83.25	HSP90AB1	Q76LV1	HS90B_BOVIN	Heat shock protein HSP 90-beta	82.0-84.0 kDa
84.72	HSP90AA1	A0A452DI85	A0A452DI85_BOVIN	Heat shock protein 90 alpha family class A member 1	84.0-86.0 kDa
84.73	HSP90AA1	Q76LV2	HS90A_BOVIN	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
84.77	HSP90AA1	A0AAF7AHH4	A0AAF7AHH4_BOVIN	Heat shock protein 90 alpha family class A member 1	84.0-86.0 kDa
86.42	TFRC	E1BIG6	E1BIG6_BOVIN	Transferrin receptor protein 1	86.0-88.0 kDa
89.86	TFRC	A0AAA9U1C1	A0AAA9U1C1_BOVIN	Transferrin receptor protein 1	88.0-90.0 kDa
93.98	EEF2	A0ABI0P2C2	A0ABI0P2C2_BOVIN	Elongation factor 2	92.0-94.0 kDa
95.37	EEF2	Q3SYU2	EF2_BOVIN	Elongation factor 2	94.0-96.0 kDa
109.83	ATP1A1	A0A3Q1MDM5	A0A3Q1MDM5_BOVIN	Sodium/potassium-transporting ATPase subunit alpha	108.0-110.0 kDa
109.92	VCL	F1N789	F1N789_BOVIN	Vinculin	108.0-110.0 kDa
111.15	VCL	A0A3Q1LIB4	A0A3Q1LIB4_BOVIN	Vinculin	110.0-112.0 kDa
112.64	ATP1A1	Q08DA1	AT1A1_BOVIN	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
112.71	ATP1A1	A0AAA9RXW7	A0AAA9RXW7_BOVIN	Sodium/potassium-transporting ATPase subunit alpha	112.0-114.0 kDa
112.97	ATP1A1	F1MI32	F1MI32_BOVIN	Sodium/potassium-transporting ATPase subunit alpha	112.0-114.0 kDa
115.73	ATP1A1	A0A3Q1LUJ8	A0A3Q1LUJ8_BOVIN	Sodium/potassium-transporting ATPase subunit alpha	114.0-116.0 kDa
116.72	VCL	A0ABI0P2J8	A0ABI0P2J8_BOVIN	Vinculin	116.0-118.0 kDa
121.98	VCL	A0ABI0P303	A0ABI0P303_BOVIN	Vinculin	120.0-122.0 kDa
123.86	VCL	A0ABI0NJJ0	A0ABI0NJJ0_BOVIN	Vinculin	122.0-124.0 kDa
129.07	VCL	A0ABI0NNT6	A0ABI0NNT6_BOVIN	Vinculin	128.0-130.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	14	0	0		routine	MGMT, . . . , HOOK3, . . . , NUDT7, , MXRA7, . . .
2.0-4.0 kDa	57	0	0	Actin family; Mitochondrial markers; Tubulin family	routine	 , RNF165, . . .
4.0-6.0 kDa	118	0	0	Actin family; Heat shock proteins; Mitochondrial markers	routine	, MNT, , DEFB12, PRM1, OST4, UBL5, DEFB1, ZNF599, KALRN, SMAGP, DEFB8, PDAP1, MSMB, PTMA, LOC132343858, DEFB13, PCD55
6.0-8.0 kDa	303	0	4	AKT pathway; Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	CYB561D1, CAMLG, MAL, MT2A, TOMM5 [Mitochondrial markers], ADH5, CYFIP1 [Actin family], DEFB9, PLN, LOC613358, RELN [NF-kB/Rel], TXNDC12, PMAIP1, PLN, , SUMO2, ZG16B, LOC768319
8.0-10.0 kDa	455	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	POLR2F, , CCDC179, UQC66 [Mitochondrial markers], SMLR1, PKIA, LOC107131456, , SPAAR, KLF2, , SMIM45, UBE2C, GTF2H5, GNG12, BBIP1 [Actin family], ATP6V1G1, TMEM167A
10.0-12.0 kDa	737	1	6	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family	elevated	, THAP6, OTOS, , SCN4B, LOC132342861, CENPW, LOC132343303, DBI, TBCA [Tubulin family], TAF13, ACBD7, , BCL2L12 [Apoptosis caspase/BCL], KCNCN, , DBI, NHP2
12.0-14.0 kDa	814	0	12	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	elevated	ADM5, DNAL4, , UBE2V2, RNF4, , PSENEN, EMC6, TANGO2, WDR83OS, , CEP20, CHCHD7, CLP5, COX5A [Mitochondrial markers], BEX3, GHRH, ,
14.0-16.0 kDa	1001	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	, , TMEM262, , LOC100847874, , DYNLRB1, , C19H17orf49, , H2AJ [Histone family], CFP468, , CDKN2B [Cell-cycle CDK/cyclin], MAX, LOC132343308, SNCA
16.0-18.0 kDa	1108	1	19	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	PLA2G2E, PLA2G1B, QDPR, COTL1, JPT1, , C1D, LOC100296105, LAMTOR3, SF3B1, MRPL27, , ELK3, , RAP1B, , MRPL27, FKBP11
18.0-20.0 kDa	1052	1	23	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RAB7B, NCBP2, ABCC9, MFAF5, UBE2I, CIRBP, RFESD, SAMD5, LRCOL1, ZBTB8OS, DCTN3 [Actin family], YY1, ALOX5AP, IER3, RAB18, SREK1P1, TAF42, NAMPT
20.0-22.0 kDa	1280	2	30	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LOC100847317, C19H17orf49, LOC125A26 [Mitochondrial markers], CDKN2AIPNL [Actin family; Cell-cycle CDK/cyclin], LOC514011, GH1, IL1RN, CSN3, MORN5, EPO, DUT, AP1S3, JDP2, IL7, MECF2, PFDN1, LOC104970886, CATHL2
22.0-24.0 kDa	1295	0	23	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	BMERB1, COQ10B [Mitochondrial markers], SAMD5, LOC100298530, ARFIP1 [Actin family], RHOC, LOC132344423, CLDN11, SAT2, PEXT [Lamin family], FHL1, PPIF [Mitochondrial markers], TRIR [Actin family], CHMP1B, C17H5orf52, RAB1B, RNASEH1, MGMT
24.0-26.0 kDa	1275	0	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	INSL6, APOO, NME5, CRIP3, , LOC781977, NUDF89, PDGFA, NAT9 [Tubulin family], DPT, CD9, LHFP3L, TATDN1, FAAP24, TMEM114, ADAMTSL1, VT11A, C15H11orf52
26.0-28.0 kDa	1242	1	32	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UBTD1, DIABLO [Mitochondrial markers], C8H9orf152, EMC10, LDHC, CCDC124, TSPAN13, MPV17L2 [Mitochondrial markers], FAM76A, MRSP18A, ELSBPB1, KCTD11, SCN1B, CSN152, MT-CO2 [Mitochondrial markers], , TM4SF1, RND1
28.0-30.0 kDa	1173	2	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	LRRC3, SINHCAF, PRP4 [Actin family], SPINT2, CMBL, LOC505326, , LRRC61, PDC2L, IL1A, FAM3B, PACRGL, YIPF5, SCAMP5, RWDD1, UCHL1, STXB86, PMP6A1
30.0-32.0 kDa	1239	6	25	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	ANXA8L1, SHFL, ARF3, TNFRSF13B, , PSMB7, THAP1, , TATDN3, SPATS1, LENG1, PLP1, POLR3F, PRSS33, RPL7L1, MS4A10, ATP5CKMT, RPL7A
32.0-34.0 kDa	1365	0	38	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	MRRF [Cell-cycle CDK/cyclin; Mitochondrial markers], BOLA-DQB, ZNF691, AGPAT1, NKX2-6, PGAM5 [Mitochondrial markers], KDELR3, CD47, DTW2D, SOD2 [Mitochondrial markers], MESD, COPE, SLA2, PDCD1LG2, SAMD10, SAP30BP, SLC25A21 [Mitochondrial markers], TRIM52
34.0-36.0 kDa	2084	5	39	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	LOC526989, NIT1, SYP, SGTA, MS4A1, IGFBP2, ZFP36L1, VTA1, OR5B32, MYG1, DHRS1, GPM6B, CDC69, CDK1 [Cell-cycle CDK/cyclin], PIM3, TMX2, SEC13, TAL1
36.0-38.0 kDa	1515	4	40	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	high	PLCXD1, HLF, TMEM231, MSR83, HNRNPA2B1, NDEL1, HNRNPAB, HNRNPA2B1, PHYH, LOC132342121, ORC67G, C1QTNF7, STBD1, ASE7, FOXR2, AUH, C19H17orf58, LDHC
38.0-40.0 kDa	1422	0	54	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	NTM, LCMT1, EFNB1, REM2, DTNBP1, AMT, ELAVL2, SCAMP1, SLX1A, TMEM268, OTULIN, YOD1, RDH16, HAPLN3, PSMD4, SGK2, GMPR2, PAQR5
40.0-42.0 kDa	1328	3	51	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	CD44, , PGA, IRF2, SLC35C1, ACKR4, ST3GAL2, ADORA2B, ZDHHC12, TIMM50 [Mitochondrial markers], GMPR2, ERI1, PPHLN1, PCBP4, HRH2, NELLE, RBM48, UBXXN1
42.0-44.0 kDa	1266	3	47	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	RPUSD4 [Mitochondrial markers], DTWD1, DHODH [Mitochondrial markers], ELF3, PEX26, , GARIN2, PTX3, ACTA2 [Actin family], ZAR1, LOC1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
68.0-70.0 kDa	686	2	25	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	PLK1, BCAM, TC7F2, AKAP8L, EPB42, BTRC, MCOLN2, ZNF792, SSX2IP [Actin family], DET1, CFI, LMBRD1 [Lamin family], C10H14orf39, LOC112442728, GNL3L, SLC6A6, KNG1, LRRC27
70.0-72.0 kDa	690	5	16	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers	high	ZWILCH, ZDHHC8, OLFML2A, ENG, ATG16L1, EYA4, MTM1, SEMA7A, FUBP1, SASS6, RLJM, KHLH31, FNBP1L, MED26, SLC6A4, TKT12, ACOT11, DMWD
72.0-74.0 kDa	598	3	13	Actin family; Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	IRAK2, LRCH4, FBXL18, XRC66, PTH1R, EPB41L4A, SOX13, SLC6A14, SH2B1, DDX43, LOC112445770, PLPPR4, HNRNPFM, PRKCG, KIFBP, FUBP1, ODA3, RAB11FIP4 [Actin family]
74.0-76.0 kDa	606	1	16	Actin family; Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	LARP4, RNF6, ACSL1, KHLH20, LEO1, TM9SF4, BB57, EFHC1, GIMAP8, XPNPEP1, ASB12, MEPC6, FZD6, ABLIM3 [Actin family], SLC27A1, RARS1, CD44, GALNT12
76.0-78.0 kDa	544	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	BRF1, SHCBP1, KATNB1, TCTN2, ZNF569, NAALAD2, CLIP4, DHX35, ZNF219, CNGA2, FCHSD1, MARCHF7, SLC44A1, HEATR3, ACSL5, BCNT2, SH3KBP1, SLC43A2
78.0-80.0 kDa	571	1	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	GAD1, SH3KBP1, RANBP9, BRD7, ARNT2, SCML2, TBC1D14, FSHR, RPH3A, APEH [Lamin family], ACSL6, ZP2, LRP8, LRRFIP1 [Actin family], MCCC1 [Mitochondrial markers], RASGRP3, SRPK1, CPSF3
80.0-82.0 kDa	561	1	11	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	elevated	IFT80, ALDH16A1, PAMR1, SLC44A2, ZER1, GPAT2 [Mitochondrial markers], DNAJC14, SEMA4F, MYH2, MTA1, PAMR1, YME1L1, ATXN7L2, DDX27, PRKCG, FAM13A, MPHOSPH10, RUFY2
82.0-84.0 kDa	503	3	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers	high	RP56KA1, LPO, DENND2B, RNF214, TRAF4, ZC3H14, DNMT3B, DNMT3B, ARNT, FGFR1, LRSAM1, PUM3, SP3, CPXM2, CDH7, MAP7, XRRRA1, RAP1GA2P
84.0-86.0 kDa	508	3	18	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	PIK3R6, SNAPP91, MID2, IQSEC3, ADD1, FCHSD2, KIAA0319L, AMFR, HSD17B4, LOC515755, ARHGAP10, CAPN11, NCSTN, CUL4B, LOXL4, CNNM4, SYTL5, DRC1
86.0-88.0 kDa	461	1	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	HSPA14 [Heat shock proteins], ELAC2, VP551, ZNF667, CHPF2, POT1, SMURF1, LOC112445144, ARHGAP10, ZNF184, GFM2 [Mitochondrial markers], ACAP1, LOC515755, EZH1 [Histone family], LOC518612, IDUA, CUX1, PCSK7
88.0-90.0 kDa	429	1	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	CDH11, ACQ2 [Mitochondrial markers], DNMT3B, TPC3, PDZRN3, ITGB5, PTPN12, FAM193B, AFG3L2, PLA2G6, XRRAL, FOSD23, MFSD6, SH3RF3, ITGB5, CCHP [Cell-cycle CDK/cyclin], LOC505600, HECTD2
90.0-92.0 kDa	419	0	14	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; p53 pathway	elevated	RINT1 [Actin family], PALD1, MAP7D1, ESPN, PTPRA, GOLGA1, OPHN1, TPCN2, GCFC2, CDH16, AFG3L2, NUP93, PBXIP1 [Actin family], ZFYVE1, PDCD6IP [Actin family], PPARGC1A [Nuclear receptor PPAR], HSPA4L [Heat shock proteins], SLC9A4
92.0-94.0 kDa	418	1	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR	elevated	IQCA1, ATG9A, NAA35, DLG1, NLGN1, TJP3, NKRIF [NF-kB/Rel], DIAPH1, IQUB, SNX25, ADGRG1, ACAP2, OSCPL11, VAV1, THSD1, ADAM23, ST14, MAP7D1
94.0-96.0 kDa	405	1	17	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	elevated	LRRC8A, PYGL, ACE2, EZF8, ATP2C2, USP16, TAX1BP1, BRINP2, CLCN2, EXD3, DNM3, PAM, FAM91A1, DNMT3B, SMPD4, CC2D1B, DSC2, KAT2A [Histone family]
96.0-98.0 kDa	400	0	17	Actin family; Heat shock proteins; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	elevated	KCNQ2, SEMA6C, TLR4, VSIG10L, TLR4, AMPD2, GCC1, PPFIBP2, TFIP11 [Actin family], MAP4K2 [MAPK cascade], HK1, MLXIPL [Actin family], KAZN, ADGRG1, , RPS6KA6, TGFBR3, TFIP11 [Actin family]
98.0-100.0 kDa	374	0	14	Actin family; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	elevated	TLR5, KCNQ4, ASAP3, ATP6V0A2, OXR1, SUCO, LRRRC8D, DNM2, VAV1, DNM3, FGD4, CHRDR, DDR2, PIK3R5, AGBL2, GTF3C3, ETTAA1, MASTL
100.0-102.0 kDa	329	0	13	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	ESYT2, TNPO1, PALM2, FAM111B, TBC1D30, DPP9, ECT2, CHSY3, AGBL2, CBL, KIF20A, RECQL5, GSAP, PIDD1 [p53 pathway], SPICE1, MAP10, DLG1, LARS2
102.0-104.0 kDa	329	0	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	elevated	ZFR2, ACTN1 [Actin family], CASD1, GALNT5, CHRDR, HACE1, EXOC1, HAUS6, AARS2, MCTP2, AASS [Mitochondrial markers], CNKSR2, ESF1, COPB2, LPIN2, MCTP2, FGD1, RNF40
104.0-106.0 kDa	304	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	LTBP3, DTHD1, CD22, MERTK, PIK3C3, TRAK1, FBXO10, ICAM5, SMARCAL1, ZNF316, TNPO3, OSBPL6, CDCCL86, SLC8A1, ACTN4 [Actin family], PKN1, ATP2C1, SMCOR8
106.0-108.0 kDa	275	0	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; Tubulin family	elevated	VP535L, TRAK1, IL6ST, ADGRD2, GRIN1, CEP120, BCLAF1 [Apoptosis caspase/BCL], EPHB6, CD163L1, STON2, ADGRF3, NEDD4L, PUM2, IPO13, MEIOC, PAM, ENPP2, RBBP6
108.0-110.0 kDa	286	2	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	FSBP, KIF28, EMILIN1, NNT, INTS4, DLG2, SLC8A1, RBM44, CORIN, INTS5, PCDHGC3, OGDHL [Mitochondrial markers], OFD1, TMEM63A, VAV3, USP37, KIF19, POLR1B
110.0-112.0 kDa	277	1	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; p53 pathway	elevated	RAI14, SNX14, KIF5C, LIMCH1, CACNA2D4, EPB41L2, ZSWIM4, MIB1, FGD1, ZBTB11, VPS18, ADGRG2, TSHZ2, EIF4ENIF1, TRIM37, RAB3GAP1, RAPGEF3, ALS2CL
112.0-114.0 kDa	277	3	5	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers	elevated	SAFB2, ANKRD35, USP38, , WDR59, EPHA7, ANKRD28, MTREX, RBM26, GRID1, CPED1, MFHAS1, PLD1, HEPH, PITRM1, ZNF628, ZCCHC14, EML4
114.0-116.0 kDa	249	1	10	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	elevated	AMOT, RASIP1 [Actin family], C2CD5, SORBS2, SENP7, EPHA5, ABL2, DDB1, PXN, HERC6, DAG1, USP28, ZNF154, LOC112442305, NLRP9, ZRANB3, CNKSR2, SLC4A4
116.0-118.0 kDa	260	1	7	AKT pathway; Actin family; Apoptosis caspase/BCL; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	BICRAL, TMEM132E, RFC1, KIF1C, STK31, FRMD4A, ADCY2, PLCH1, TAOK1, PLEKHA7, LOC100297905, XPO1, ATP8A2, EMC1, PPARGC1B [Nuclear receptor PPAR], IDE, CDCC57, MCM9
118.0-120.0 kDa	191	0	8	Actin family; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	elevated	MCF2L, MED24, ZNF445, RPS6KC1, LRIG3, ATP9A, RAPGEF4, IDE, PTK7, HDAC4 [Histone family], TT11 [Actin family], IDE, POLD1, PPP1R13B, FBRSL1, NLRP3, SLC4A7, CPED1
120.0-122.0 kDa	233	1	2	AKT pathway; Actin family; Apoptosis caspase/BCL; Histone family; MAPK cascade; p53 pathway	elevated	CARD10 [Apoptosis caspase/BCL], SEMA58, DAG1, ROR2, GPR89A, C2CD5, FBXW10B, SEC24A, LARS1, FBH1, MTS18BP1, HDAC5 [Histone family], LRIG2, CEP128, MYRF, SYNPO, FAM120C, TNKS2
122.0-124.0 kDa	203	1	4	Actin family; JAK-STAT; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	GRIP1 [Actin family], MEGF10, BNC2, PLEKHA7, LIMCH1, WDR6, SEMA5B, SMURF2, CORIN, TCERG1, SLC12A6, CEBP2, PLCB1, DLC1, STARD8, POLR1B, FMNL2, UNC13D
124.0-126.0 kDa	209	0	4	Actin family; Histone family; Tubulin family; p53 pathway	elevated	IQSEC1, ANO8, MICAL2 [Actin family], PPIP5K2, ADAR, MAN2A1, PDGFRB, SRGAP1, PLD1, FBLN2, SPECC1L, GRIP1 [Actin family], PHF2, AP3B2, MTR, , SCAF4, AFF4
126.0-128.0 kDa	180	0	6	Actin family; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	CFAP70, TBC1D8B, FBF1, RANBP17, TPP2, ATP8A2, AHI1, PKP4, EFL1, GRM5, MYO1E, SASH1, TAOK2, ADGRL2, LAT51, POLRMT, USP7, HIPK1
128.0-130.0 kDa	185	1	6	Actin family; JAK-STAT; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	HIPK2, MYBPCL, AP4E1, ADGRG6, FAM184A, ATP2B2, RANBP17, FNIPI [Actin family], ERC1 [Actin family], PLEKHN1, RBL2, ZNF592, DAB2IP [Actin family], CEP128, USP43, MCF2L, SIN3B, MCF2L
130.0-132.0 kDa	184	0	6	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Lamin family; Tubulin family; p53 pathway	elevated	ZBTB40, PUM1, ATP8A1, MBD1, ITSN1, UBE4B, RPTN, KCNMA1, CAMTA2, RBM20, ADCY6, SMC3, NRCAM, PPP2R3A, PTPN14, PPFIA1 [Actin family], MYO1E, KCNMA1
132.0-134.0 kDa	193	0	4	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	COBLL1, BRD1, CILP, TTF2, MTMR3, TBC1D8, TBCD [Tubulin family], RPTOR, PLCB1, CATSPERG, TBC1D1, ZC3H4, DAB2IP [Actin family], PLEKHA6, MEI1, PPFIA2 [Actin family], POLR2B, SMC7
134.0-136.0 kDa	162	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Tubulin family	elevated	SASH1, CTC1, ATP2B2, EGFR, LOC515570, DGKD, IQSEC1, IGDC04, LOC516849, CCNF [Cell-cycle CDK/cyclin], ATP11B, TBCD [Tubulin family], POLG, ERBIN, LTBP3, ADCY6, LTBP3, RTEL1
136.0-138.0 kDa	154	0	0	Actin family; Histone family; NF-kB/Rel	routine	NID1, DHX30, SMC1B, ARHGEF2, RAPH1, MECOM [Histone family], CEP128, PCDH9, ITPRID2 [Actin family], IRS2, CGN, EHBP1, PLCB4, LOC100847574, SEC31A, ABCB11, ADGRG6, SUCO
138.0-140.0 kDa	166	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Tubulin family	elevated	SALL1, TAF2, KIF168, MECOM [Histone family], GAK [Cell-cycle CDK/cyclin], TTC17, POLRMT, SYNRG, AP3D1, KANK1, RTL9, MTUS1, ALPK1, FLII, GNPTAB, AUTS2, ULK4, ITPRID2 [Actin family]
140.0-142.0 kDa	152	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; p53 pathway	elevated	ADCY8, PFAS, RIPOR1 [Actin family], TIMELESS, KIF4A, MTR, MTERF4, STAG1, AOX1, DENND3, JCAD, TASOR2, FAM120A, CRAMP1, WDR17, OVCH1, RAPGEF1, WDR35
142.0-144.0 kDa	128	0	2	Actin family; Histone family; MAPK cascade	elevated	PLCG2, ASXL2, ZMYM3, PLEKHA7, TNKS, MAP3K6 [MAPK cascade], ATP13A3, DNAAF9, SORBS1, MMS22L, PEX1, TBC1D1, PHLPP1, LOC100848700, PPFIA2 [Actin family], SYNJ2, PER2, TRPS1
144.0-146.0 kDa	128	0	7	Actin family; Cell-cycle CDK/cyclin; MAPK cascade; Mitochondrial markers	elevated	PNPLA6, ERBB4, APAF1, SK3, FRMPD4, TECPR2, RNF123, CENP3, PHLPP2, NIN, KDR, NFASC, PAR3, RASAL2, LOC100337390, VWASB, SFI1, SMC1A
146.0-148.0 kDa	116	0	3	Actin family; Histone family; Lamin family; MAPK cascade	elevated	SYNRG, LOC530599 [Lamin family], WDR17, SYNRG, CD109, SPAG9 [Actin family], UBN2, , TDRD9, CDHR2, SPAG5, UBN2, ZNF532, ADGRL2, RNF123, ELP1, STK36, RGSG22
148.0-150.0 kDa	120	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade	elevated	PNPLA6, CNTLN, NUP155, ABCC4, MYO6, RIPOR1 [Actin family], PLB1, LOC528412, RIC1, AOX4, CADPS2, CARMIL2, SOS2, ZNF423, DENND3, AOX2, PLEKHA7, TMEM94
150.0-152.0 kDa	103	0	1	Actin family; Histone family	routine	MAGI1, ACE, CECR2, CARMIL3, RGS12, PHLPP2, ABCB11, COL17A1, KIF27, KTN1, ARHGEF18, TONSL, MOV10L1, TTC41, RAB3GAP2, USP42, USP19, FYCO1
152.0-154.0 kDa	105	0	2	Actin family; Cell-cycle CDK/cyclin; MAPK cascade; Mitochondrial markers	elevated	MYBBP1A, LOC508459, NLRP1, LOC508459, UBE4B, FRMPD2, RAB3GAP2, ZFYVE9, KIF24, TRIO, RPRD2, RIMS1, TMEM94, KIF27, RAB3GAP2, CPS1 [Mitochondrial markers], CD109, DAB2
154.0-156.0 kDa	89	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	ZMYM2, CNGB1, PTPRG, AKNA, COL4A1, TONSL, , KDM6A [Histone family], ABCC11, RAB3GAP2, LRPPRC [Mitochondrial markers], SBN01, ROCK1, HEG1, ATP7A, ADGRB2, LOC100337390, CLIP1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
156.0-158.0 kDa	87	0	1	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers	routine	COL4A4, MAP4K4, MEGF6, KIF24, RPRD2, GOLGA3, FANCD2, RAB3GAP2, ARAP1, GPATCH8, ABCC5, ROBO2, LRPRRC [Mitochondrial markers], CLIP1, DISP1, MAGI1, GPATCH8, MAP4K4
158.0-160.0 kDa	76	0	4	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers	elevated	NCAPD3, CPSF1, SCAPER [Cell-cycle CDK/cyclin], PLEKHG1, FANCD2, LMTK2, PRG4, WDR62, DCC, PTPRG, NPHP4, TECPR2, GRIN2A, ROCK1, DMBT1, DMBT1, NWD1, DMBT1
160.0-162.0 kDa	127	0	4	Actin family; Apoptosis caspase/BCL; Histone family	elevated	RASAL2, RIMBP2, PIPISK1, CLASP2, COL4A6, MED14, COL22A1, NHS [Actin family], GTF2H2, ATP10B, TDRD9, COL4A1, BLM, CCDC141, YTHDC2, ARHGAP23, NRXN1, ROCK2
162.0-164.0 kDa	79	0	3	Actin family; Histone family; MAPK cascade	elevated	FANCD2, MLH3, DOT11 [Histone family], , FLT1, ARHGAP23, ADGRL1, ADGRL1, UGGT2, CUX2, SCAF8, SHROOM4, AZML1, PTPRU, COL4A6, SHROOM2, CTTNBP2 [Actin family], EPRS1
164.0-166.0 kDa	86	0	5	AKT pathway; Actin family; Cell-cycle CDK/cyclin; MAPK cascade; Mitochondrial markers	elevated	RIMS1, CDK12 [Cell-cycle CDK/cyclin], PTPRK, ROCK2, WDR62, CLASP2, COL4A4, COL4A6, FRMPD1, PXDN, FYCO1, NHS [Actin family], PLEKHG1, DIP2C [Actin family], TOPBP1, MRC1, COL16A1, RSF1
166.0-168.0 kDa	78	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	CCDC18, PHLPP1, NISCH, FGD5, MRC2, SCRIB, TMEM131L, SCRIB, ADGRB2, MEGF6, LOC506828, ATAD2B, ZNF518A, A2M, NAIP, A2M, A2M, TEX14
168.0-170.0 kDa	70	0	1	Actin family; Histone family; Lamin family; p53 pathway	routine	COL11A1, KDM5B [Histone family], ABCC2, , ADGRB3, ADGRL3, IQSEC2, TRPM1, NHL1, FLNC [Lamin family], , PIK3C2B, POLA1, RAPGEF6, KDM5B [Histone family], PLEKH2, GEMIN5, ATAD2B
170.0-172.0 kDa	68	0	2	AKT pathway; Actin family; Histone family; Lamin family; Tubulin family	routine	RERE, NHL1, SHROOM2, PIK3C2G, ARHGEF12, CEP170B, FHAD1, ARHGEF12, DIP2A [Actin family], DOT1L [Histone family], COL5A3, PIK3C2G, ADGRL3, KIAA0586, SCRIB, ALK, DEPDC5, ARHGAP35
172.0-174.0 kDa	64	0	2	Actin family; Histone family; MAPK cascade; Nuclear receptor PPAR	routine	RAPGEF2, FAM135A, PLXNC1, ARHGAP5, COL5A3, RAPGEF6, CFTR, COL11A2, ARHGAP5, COL11A2, DEPDC5, ZMYM4, LRRCT, DISP1, DIP2B [Actin family], MEGF6, INO80, ZMYM4
174.0-176.0 kDa	44	0	2	Actin family; MAPK cascade; Tubulin family	routine	MEGF6, AGL, , ABCG9, KIF21B, TOP2A, PTPN23, TMEM131L, PHRF1, DIP2C [Actin family], TMEM131L, ABCC9, LRP6, TMEM131L, RAPGEF6, ATRN [Actin family], RANBP2, MADD
176.0-178.0 kDa	59	0	0	Actin family; Lamin family; Nuclear receptor PPAR	routine	ADGRL3, UGGT1, ALK, MBD5, SMG6, TUT7, THSD7B, CEP170, SKIC3, ERCC6L2, PTPN23, CEP170, ALS2, LRP6, ROBO1, KDM6B, HECW2, EIF4G1
178.0-180.0 kDa	46	0	3	Actin family; Histone family; MAPK cascade; Tubulin family	routine	DNMBP, CTTNBP2 [Actin family], PIK3C2B, DNMBP, KIF26B, MEGF6, C3, ABCA9, KIF21B, TUT7, IQGAP2, ADAMTS12, IQGAP2, SYNM, TUBGCP6 [Tubulin family], PLXNC1, KIF21A, EIF4G3
180.0-182.0 kDa	58	0	2	Actin family	routine	LTBP2, RIMS2, , TOP2B, KATNIP [Actin family], LRP5, N4BP2, UNC13B, KIF21A, USP54, KIF21B, TRPM3, C3, EIF2AK4, LMO7, CC2D2A, RNF17, USP54
182.0-184.0 kDa	48	0	0	Actin family; Histone family	routine	MON2, PRDM2, AFDN, CEP170, MROH1, DOT1L [Histone family], WNK3, UNC13B, RNF17, SMARCA4 [Actin family], ITSN2, MON2, CC2D2A, MARF1, COL11A1, TARBP1, PTPRV, MADD
184.0-186.0 kDa	55	0	1	Actin family; Histone family	routine	DNMT1, MBD5, PIK3C2B, PBRM1, AKAP12, LRP6, CEP290, ABCA6, RAPGEF2, UNC13B, POLR1A, KIF1B, ABCA3, ADAMTSL3, TET3, RESF1, IQGAP3, MON2
186.0-188.0 kDa	51	0	0	Lamin family	routine	ABCA5, CAMSAP1, KDM5A, CC2D2A, IQGAP3, KDM5A, THOC2, IQGAP1, LAMB1 [Lamin family], LOC528040, MROH2A, ABCA16, KALRN, SPHKAP, TRPM1, LAMB4 [Lamin family], ZSWIM8, RREB1
188.0-190.0 kDa	39	0	0	Actin family	routine	RNF17, PLCH1, UBR2, ABCA14, RICTOR, RICTOR, ARID2, SIPA1L2, PLCH1, NRXN2, TNRC6A, C5, KIAA1671, CTTNBP2 [Actin family], KIF26A, ARID2, PBRM1, KIAA1210
190.0-192.0 kDa	43	0	1	Actin family; Histone family; Lamin family	routine	BCORL1, FRMPD3, NRAP, CCDC180, KAT6B [Histone family], MROH2A, GCC2, NRAP, RICTOR, MGAM, ANKRD26, MON2, KALRN, TOGARAM1, GCC2, MON2, PARP14, PARP4
192.0-194.0 kDa	67	0	0	p53 pathway	routine	TNRC6B, MROH1, BTBD8, CLTC, MARF1, CHD1, TNS1, TRIP12, KALRN, KIF1A, SYTL2, ARHGEF28, KDM3B, SIPA1L2, CLTC, ABCA3, ADAMTSL1, BRWD3
194.0-196.0 kDa	50	0	0	Actin family; Lamin family	routine	PCDH15, SPEEF2, POLR1A, PBRM1, NRAP, ZSWIM8, KIDINS220 [Actin family], SIPA1L3, PBRM1, SBF1, ARAP2, ARHGEF28, TNS1, WDR90, DNMT1, ARID2, ZC3H13, PTPRF
196.0-198.0 kDa	50	0	1	Actin family; Lamin family	routine	LAMB2 [Lamin family], DNMT1, MICAL3 [Actin family], QSER1, KIDINS220 [Actin family], ARFGEF2, WDR90, NLRC5, GBF1, PTPRF, QRICH2, LAMA4 [Lamin family], PARP4, NUP188, SKIC3, ARAP2, CEP152, ANKRD26
198.0-200.0 kDa	50	0	3	Actin family; Cell-cycle CDK/cyclin; Lamin family	routine	ZSWIM8, RAI1, NAV3, BRWD3, CACNA1I, MGAM, TJPI, PCDH15, TNS1, UBR2, ZNF106, BRWD3, MIA3, LAMA4 [Lamin family], LTBP2, LMO7, N4BP2, WDR90
200.0-202.0 kDa	51	0	1	Histone family	routine	KIF1A, TSC2, SHANK2, BRCA1, PTPRF, TANC1, KIF16B, NALCN, KMT2E [Histone family], CCDC1, LMO7, ITGB4, ANK1, NAV1, TOGARAM1, ITGB4, UBR2, UBR2
202.0-204.0 kDa	35	0	1	Cell-cycle CDK/cyclin; Lamin family; p53 pathway	routine	TNS1, ZNF106, LCOLR, CPAMD8, TNRC6A, KIAA1217, VIRMA, MAP2, LMO7, MGAM, BRWD3, NCKAP5, LAMA4 [Lamin family], SHPRH, PARP4, PARP14, PSME4, MYO5C
204.0-206.0 kDa	29	0	0	Actin family; Histone family	routine	FOCAD, HELZ, ZNF142, LTN1, TMEM131, SHANK2, KIF1B, KIF26B, PATJ, KMT2E [Histone family], PHIP [Actin family], WIZ, NCOA6, NLRC5, CCDC88A, ARHGEF4, RALGAP2, ZNF236
206.0-208.0 kDa	47	0	1	Actin family; p53 pathway	routine	WDR90, PLXNA4, AFDN, HEATR5B, ARFGEF2, TJP1, ARFGEF1, BRCA1, BRCA1, SCN11A, SHROOM3, FNDC1, HCFCL, KIF1B, PCSK5, MDC1, AKAP11, CHD2
208.0-210.0 kDa	48	0	2	Actin family; Cell-cycle CDK/cyclin; p53 pathway	routine	CCDC1, CDK5RAP2 [Cell-cycle CDK/cyclin], SBF2, DICER1, ZFC3H1, NOTCH4, MGAM, KIAA1549, PLXNA3, DLG5, ZNF236, SI, PDCD11, CHD2, PDCD11, TNRC6A, ARFGEF1, PARP4
210.0-212.0 kDa	53	0	2	Cell-cycle CDK/cyclin; p53 pathway	routine	TRPM7, BAZ2A, PARP4, CDK5RAP2 [Cell-cycle CDK/cyclin], GBF1, ANAPC1, ANK3, AKAP11, VWA8, SBF1, FER1L6, DENND4C, PLXNB2, KIF13B, AKAP3, MIA3, PLXNA2, GBF1
212.0-214.0 kDa	44	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; p53 pathway	routine	PLXND1, PTPRD, FER1L6, LOC101902149, FER1L6, ARHGEF4, ANK3, SETD1B [Histone family], CDK5RAP2 [Cell-cycle CDK/cyclin], CROCC, MCM3AP, PTPRF, KIF20B, PCDH15, ATG2A, ANKRD31, PRRI14L, UBR3
214.0-216.0 kDa	37	0	0	Actin family	routine	WNK2, ANAPC1, PTPRB, MYO5A, MYO5A, MIA3, DENND4A, PTPRB, MPD2, ARFGEF1, DICER1, ANK3, PLXNA1, PTPRD, MICAL3 [Actin family], PCDH15, TTC6, PPL
216.0-218.0 kDa	42	0	1	Actin family; Cell-cycle CDK/cyclin; p53 pathway	routine	ADAMTS9, ARHGAP21 [p53 pathway], CDK5RAP2 [Cell-cycle CDK/cyclin], ANAPC1, NCOA6, MYH4, MAP2, ANKRD31, DLG5, TAF1, MYLK, AGRN, PTPRS, TTC3, DOCK4, CROCC, KIF26B, ARHGAP21 [p53 pathway]
218.0-220.0 kDa	31	0	1	Actin family; Apoptosis caspase/BCL; p53 pathway	routine	DOCK4, CKAAP5, DICER1, EML5, WNK2, MYO5A, SCN1A, DICER1, AGRN, WNK2, PTPRS, EML5, CASP8AP2 [Apoptosis caspase/BCL], YLPM1, MPD2, CRYBG1, NUP214 [p53 pathway], LCT
220.0-222.0 kDa	24	0	1	Apoptosis caspase/BCL	routine	TRIP12, BRD10, SMCHD1, AK9, SCN8A, CHD4, ARHGEF17, MYH1, TTC3, TNC, PTPRS, TNC, CASP8AP2 [Apoptosis caspase/BCL], SDC1, CRYBG1, MYH7B, CRYBG1, MYH15
222.0-224.0 kDa	44	0	1	Actin family; Apoptosis caspase/BCL; Histone family	routine	MYLK, ZFC3H1, MYH4, KIF13B, SCN5A, WNK1, DSCAM, HEATR1, TRIP12, SCN10A, MYH4, CROCC, PCNX3, SCN3A, PCNX3, MYH8, MICAL3 [Actin family], PTPRB
224.0-226.0 kDa	30	0	0	Histone family	routine	DOCK8, MED13, SCN2A, MED13L, HEATR5B, DOCK6, BRD10, SHANK1, CRYBG1, MYH11, SCN9A, FANCM, LRRK1, FER1L5, CEP20, DOCK8, CKAAP5, WNK1
226.0-228.0 kDa	27	0	0		routine	SCN5A, TET2, ZFC3H1, ESPL1, SON, SCN8A, MYO18A, SHANK1, FANCM, ARHGAP32, SMCHD1, RALGAP1A, PHF3, OTOF, PCNX2, TRIP11, MYH9, BAZ2B
228.0-230.0 kDa	33	0	1	Cell-cycle CDK/cyclin; Histone family	routine	SCN5A, BRD10, CDK5RAP2 [Cell-cycle CDK/cyclin], DOCK11, DOCK9, MYH10, MYH10, MYH11, TANC2, TRIP12, ALPK2, TRIP12, WNK1, MYH10, SCN1A, KIAA1549L, ARFGEF3, KAT6B [Histone family]
230.0-232.0 kDa	32	0	0		routine	MALRD1, NUP205, FANCM, DOCK11, MYH10, BAZ2B, FANCM, CRYBG1, MYO10, DSCAML1, MYO18A, ANKRD12, TRPM6, ARHGEF17, ESPL1, ICE1, PIKFYVE, TNC
232.0-234.0 kDa	32	0	0	Actin family	routine	PCM1, TDRD6, ICE1, MYO18A, NAV3, MYH14, P4KA, NAV3, PIKFYVE, NIN, MYH10, PLXNB1, CIT [Actin family], CCDC88A, MYOF, ATG2B, MYO18A, DOCK11
234.0-236.0 kDa	31	0	0	Actin family	routine	DOCK9, RP1, LOC101904413, ZNF638, MYLK, MYH10, ABCA7, UNC13C, DOCK11, BAZ2B, CIT [Actin family], DOCK8, HEATR1, MYO10, P4KA, , DOCK3, CABIN1
236.0-238.0 kDa	28	0	0	Actin family	routine	ARFGEF3, MALRD1, MALRD1, GTF3C1, MED12L, HEATR1, PIKFYVE, DOCK9, DOCK6, CIT [Actin family], DOCK8, MED12, DOCK9, NBAS, FRAS1, , PIKFYVE, PHF3
238.0-240.0 kDa	27	0	0		routine	SORL1, MYH14, DOPIB, PCM1, PIKFYVE, SORL1, PTPRB, DOCK7, CACNA1C, DOCK8, DYSF, TET1, SVIL, INTS1, FER1L5, MED13, DOCK11, F5
240.0-242.0 kDa	33	0	0		routine	DYSF, DYSF, SORL1, DOCK9, DOCK9, MED12, DOCK9, NBAS, DYSF, NBAS, ARFGEF3, DYSF, TNC, MED12L, PRRC2B, BAZ2B, MED12, TNC
242.0-244.0 kDa	23	0	0		routine	MYO9B, ARID1A, NBAS, CACNA1D, MED13L, CACNA1D, MALRD1, HEATR1, SVIL, DOCK7, ACAN, NIN, ZGRF1, MYO9B, DOCK3, DOCK9, INTS1, CAD
244.0-246.0 kDa	18	0	0		routine	SEC16A, LOC616782, SNRNP200, NAV3, NOTCH3, CACNA1I, MYO7B, WNK1, SVIL, SVIL, CACNA1G, FREM1, BAZ2B, CACNA1D, MYO7B, GON4L, NAV3, CAD
246.0-248.0 kDa	24	0	0		routine	CACNA1D, MALRD1, NAV3, SEC16A, ACAN, DOCK3, ARID1B, KNTC1, GON4L, GON4L, PCNX1, ACAN, CACNA1H, NIN, CACNA1H, COL6A6, DOCK3, RTTN
248.0-250.0 kDa	13	0	0		routine	CLIP1, MYO7A, PTPRB, ANKRD17, AKAP6, F5, ZNF318, DOP1B, NAV2, SON, MYO7A, NIN, DOCK10
250.0-252.0 kDa	16	0	0		routine	UNC13C, DOCK10, CSPG4, KNTC1, ascc3, TNC, SORL1, MYO7A, PKDREJ, SON, ARID1B, NCOR1, HECTD1, OTOGL, DOCK10, CLIP1
252.0-254.0 kDa	10	0	0		routine	KIAA0100, SPAG17, KIAA0100, KIAA0100, PLCE1, ARID1B, ACACB, AHCTF1, LOXHD1, AHCTF1
254.0-256.0 kDa	18	0	0	Actin family; Heat shock proteins	routine	NAV2, CACNA1G, ABCA1, ABCA1, PDE4DIP [Actin family], URB1, DNAJC13 [Heat shock proteins], C2CD3, CACNA1E, ASCC3, PTPRZ1, POLE, NAV3, DOP1B, NACAD, F8, CACNA1G, NACAD
256.0-258.0 kDa	15	0	0		routine	ZNF318, NAV2, AKAP6, ACACB, AHCTF1, PTPRQ, TTC28, GOLGA4, CIC, CIC, PLCE1, ABCA4, NAV2, HIVEP3, PCNX1
258.0-260.0 kDa	15	0	0	Actin family; Heat shock proteins; Histone family	routine	C2CD3, SPG11, ABCA2, CACNA1G, TTC28, DNAJC13 [Heat shock proteins], POLE, F8, CEP290, ACACA, OTOGL, CREBBP [Histone family], CACNA1G, C2CD3, PDE4DIP [Actin family]
260.0-262.0 kDa	14	0	0	Heat shock proteins; Histone family	routine	NCOR1, NAV2, EP300 [Histone family], LOXHD1, F8, TNC, NAV2, HECTD1, CACNA1B, CACNA1B, LOC101903126, LOC101903126, DNAJC13 [Heat shock proteins], VCAN

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
262.0-264.0 kDa	17	0	0	Histone family	routine	NACAD, ROS1, CACNA1G, POLE, SPTBN2, SPG11, ACACA, ANKRD17, SON, GPR179, BRWD1, CACNA1G, PEG3, F8, KNL1, NOTCH2, EP300 [Histone family]
264.0-266.0 kDa	20	0	0	Histone family	routine	PEG3, NOTCH2, IGF2R, ANKRD17, OTOGL, CACNA1A, TASOR2, NCOR1, PEG3, SPTBN2, GOLGA4, KIAA0100, CREBBP [Histone family], ACACA, NOTCH2, ACACA, RIF1, NOTCH1
266.0-268.0 kDa	17	0	0	Actin family	routine	CNTRL, TLN1, PRPF8, STAB2, ACACB, ZAN, NCOR2, SEH1L, DOP1A, HIVEP2, ATRX, CEP192, PDE4DIP [Actin family], PRPF8, CNOT1, SEH1L, GOLGA4
268.0-270.0 kDa	25	0	0		routine	PRPF8, POLQ, GOLGA4, NACAD, ABCA2, TLN1, FASN, GOLGA4, SPTB, RIF1, TPR, SEH1L, ZFHX2, CNTRL, TTC28, CEP290, SPTBN2, DMD
270.0-272.0 kDa	13	0	0	Actin family	routine	SPTBN2, ACACA, NCOR2, GOLGA4, IGF2R, TLN1, NBEA, FASN, PDE4DIP [Actin family], TTC28, SPTBN2, TLN1, NACAD
272.0-274.0 kDa	21	0	0	Actin family; Lamin family	routine	FN1, CUL9, MYO18A, ZFHX2, NCOR1, SPTBN1, TLN2, ATRX, ANKHD1, ACACB, CEP192, CHD9, SPTBN2, PDE4DIP [Actin family], PRPF8, NCOR2, CEP250, CUL9
274.0-276.0 kDa	16	0	0	Actin family; Lamin family	routine	MYO18A, SPTBN1, IGF2R, IGF2R, FASN, ABCA2, TPR, FN1, FLNB [Lamin family], FASN, SETX, STAB1, ANKRD17, LRRK2, PDE4DIP [Actin family], CACNA1A
276.0-278.0 kDa	14	0	0	Lamin family	routine	CEP250, PDZD2, JMJD1C, NCOR2, DOP1A, LRRK2, STAB2, FLNB [Lamin family], ATRX, LOC616782, COL6A5, DOP1A, ZFYVE26, DOP1A
278.0-280.0 kDa	12	0	0	Actin family; Lamin family	routine	FLNB [Lamin family], USP24, JMJD1C, , CEP295, MYO18B, PDE4DIP [Actin family], STAB1, RIF1, MYO18B, SPTAN1, SRRM2
280.0-282.0 kDa	23	0	0	Histone family; Lamin family	routine	CACNA1A, MYO18B, FLNA [Lamin family], TENM2, ZFYVE26, BAHCC1, NCOR2, CUL9, SPTAN1, FLNA [Lamin family], SPTA1, TENM2, CEP295, NSD1 [Histone family], CEP290, TENM2, ATRX, GCN1
282.0-284.0 kDa	14	0	1	Actin family	routine	RIF1, ZFYVE26, CACNA1A, SPG11, PDE4DIP [Actin family], COL6A5, SPTAN1, PIEZO1, COL6A5, PIEZO1, ZNF292, JMJD1C, ABCA12, MTOR
284.0-286.0 kDa	18	0	0	Histone family	routine	BDP1, USP9X, EPG5, NSD1 [Histone family], CEP290, POLQ, SPTAN1, SPTAN1, RALGAPA1, SPTAN1, FN1, RALGAPA1, SPTAN1, HIVEP1, NBEAL1, PDZD2, CEP290, FBN3
286.0-288.0 kDa	15	0	0	Histone family; Lamin family	routine	LRRK2, VWF, SPTAN1, SETD2 [Histone family], ANKRD11, TENM1, COL6A5, FLNC [Lamin family], CEP295, ANKRD11, MUC5B, ATRX, SPTAN1, TENM2, IGSF10
288.0-290.0 kDa	11	0	1	Histone family	routine	BAHCC1, KMT2B [Histone family], USP24, SPTBN4, MTOR, APC, HECTD1, GCN1, TENM2, SRRM2, POLQ
290.0-292.0 kDa	12	0	1	Lamin family	routine	ZNF462, CEP295, CEP295, MTOR, ANKRD11, FLNC [Lamin family], GCN1, CEP290, USP9X, FLNC [Lamin family], UNC79, CHD8
292.0-294.0 kDa	15	0	0	Histone family	routine	TASOR2, EPG5, GCN1, GCN1, USP9X, USP9Y, CHD9, MYO9A, SRRM2, ABCA12, COL7A1, KMT2B [Histone family], CFAP46, SRRM2, TENM2
294.0-296.0 kDa	13	0	0		routine	GCN1, USP24, SETX, ZAN, ATR, TACC2, BOD1L1, TEP1, NBEAL1, PRRC2C, PDZD2, TASOR2, CEP295
296.0-298.0 kDa	5	0	0	Histone family	routine	PRRC2C, NSD1 [Histone family], MEGF8, SRRM2, UNC79
298.0-300.0 kDa	4	0	0		routine	CRYBG3, NIPBL, ZNF292, AKAP13
300.0-302.0 kDa	12	0	0		routine	MGAM, ITPR2, FBN1, TENM3, AKAP13, TG, FBN1, UNC79, ITPR2, ITPR3, TENM3, NBEAL1
302.0-304.0 kDa	20	0	0		routine	TG, NF1, ATR, PRRC2C, ABCA2, TNRC18, MXRA5, ITPR3, AKAP13, TG, SETX, TG, AKAP13, DST, TNRC18, NF1, UNC79, ZNF292
304.0-306.0 kDa	7	0	0		routine	FBN3, ITPR1, TENM1, CHD6, AKAP13, TENM2, TENM1
306.0-308.0 kDa	13	0	0		routine	CENPE, PIEZO2, TENM4, TNRC18, PRRC2C, TENM2, TENM4, DMXL1, TNRC18, ITPR2, CRYBG3, NBEAL1,
308.0-310.0 kDa	19	0	0		routine	TNRC18, ITPR2, PRUNE2, PRUNE2, ITPR1, VWF, CHD6, TEX15, PRUNE2, TENM4, LRBA, UBR5, CENPE, APC, DMXL1, PRUNE2, NBEAL1, PRUNE2
310.0-312.0 kDa	6	0	0		routine	VWF, APC, PRUNE2, PRRC2C, , LRBA
312.0-314.0 kDa	10	0	0		routine	FBN1, UBR5, TENM4, OTOG, TNRC18, APC, NF1, MXRA5, CRYBG3, FBN1
314.0-316.0 kDa	8	0	0		routine	APC, FBN2, NF1, LOC616782, TENM4, PRRC2C, PIEZO2, APC
316.0-318.0 kDa	5	0	0		routine	TRANK1, NIPBL, PIEZO2, CELSR2, LRBA
318.0-320.0 kDa	5	0	0		routine	UTP20, FCGBP, BOD1L1, CEP290, CRYBG3
320.0-322.0 kDa	4	0	0		routine	FRY, MGA, NF1, NF1
322.0-324.0 kDa	4	0	0		routine	NF1, CEP290, DSP, MAP1A
324.0-326.0 kDa	8	0	0	Histone family	routine	FBN1, COL12A1, RANBP2, EP400, CELSR2, BPTF, PIEZO2, ASH1L [Histone family]
326.0-328.0 kDa	5	0	0		routine	CHD9, COL6A3, COL6A3, NBEA, NBEA
328.0-330.0 kDa	1	0	1		routine	SRCAP
330.0-332.0 kDa	5	0	0		routine	NBEA, COL6A3, HTT, CELSR1, BPTF
332.0-334.0 kDa	6	0	0		routine	CHD7, DSP, NF1, EP400, ZZEF1, COL12A1
334.0-336.0 kDa	6	0	1	Lamin family	routine	SRCAP, LAMA2 [Lamin family], NF1, MAST4, ZZEF1, SYNE1
336.0-338.0 kDa	3	0	0		routine	EP400, EP400, LAMA1
338.0-340.0 kDa	12	0	0		routine	MAP1A, FRY, BPTF, RANBP2, DMXL2, FRY, FRY, TRANK1, KALRN, COL6A3, MAST4, DMXL1
340.0-342.0 kDa	9	0	0		routine	DMXL2, EP400, FRYL, TRANK1, MAST4, TRANK1, MGA, FRYL, DMXL1
342.0-344.0 kDa	7	0	0	Lamin family	routine	IQCN, TRIO, DMXL2, HTT, TRIO, LAMA2 [Lamin family], FRY
344.0-346.0 kDa	2	0	0		routine	HTT, RANBP2
346.0-348.0 kDa	3	0	0		routine	DCHS1, CEP350, TRIO
348.0-350.0 kDa	4	0	0		routine	VPS13A, WDR87, FRYL, ATM
350.0-352.0 kDa	7	0	0		routine	ATM, VPS13A, DNAH12, CEP350, FREM2, FRYL, FRY
352.0-354.0 kDa	5	0	0		routine	CFAP54, MAST4, REV3L, WDFY4, PCNT
354.0-356.0 kDa	6	0	0		routine	SPEG, MAST4, PCNT, SPEG, DCHS1, LOC101902689
356.0-358.0 kDa	3	0	0		routine	PCNT, CFAP47, MKI67
358.0-360.0 kDa	7	0	0	Lamin family	routine	CPLANE1, LAMA3 [Lamin family], MAST4, CELSR3, PCNT, CDH23, CFAP54
360.0-362.0 kDa	4	0	0	Lamin family	routine	VPS13A, UNC80, SZT2, LAMA3 [Lamin family]
362.0-364.0 kDa	2	0	0		routine	ATM, UNC80
364.0-366.0 kDa	4	0	0	Lamin family	routine	CENPF, CPLANE1, MKI67, LAMA5 [Lamin family]
366.0-368.0 kDa	5	0	0	Lamin family	routine	LAMA5 [Lamin family], GOLGB1, LAMA3 [Lamin family], LOC539323, GOLGB1
368.0-370.0 kDa	6	0	0		routine	GOLGB1, SVEP1, VCAN, CSMD1, MKI67, VCAN
370.0-372.0 kDa	4	0	0		routine	VCAN, UNC80, GOLGB1, MKI67
372.0-374.0 kDa	2	0	0		routine	GOLGB1, GOLGB1
374.0-376.0 kDa	3	0	1	NF-kB/Rel	routine	CENPF, RELN [NF-kB/Rel], EPPK1
376.0-378.0 kDa	2	0	0		routine	BRCA2, GOLGB1
378.0-380.0 kDa	2	0	0		routine	CENPF, CSMD2
384.0-386.0 kDa	1	0	0		routine	BRCA2
386.0-388.0 kDa	3	0	0		routine	CSMD1, CSMD3, CSMD1
388.0-390.0 kDa	4	0	3	NF-kB/Rel	routine	SVEP1, RELN [NF-kB/Rel], RELN [NF-kB/Rel], RELN [NF-kB/Rel]
390.0-392.0 kDa	4	0	0		routine	CSMD2, WDFY3, UTRN, CUBN
392.0-394.0 kDa	3	0	0		routine	WDFY3, UTRN, UTRN
394.0-396.0 kDa	7	0	0		routine	USP34, CSMD3, UTRN, UTRN, ASPM, CSMD2, ZFHX4
396.0-398.0 kDa	2	0	0		routine	CUBN, ZNF469
398.0-400.0 kDa	1	0	0	Lamin family	routine	LAMA5 [Lamin family]
400.0-402.0 kDa	1	0	0	Actin family	routine	SPEN [Actin family]
402.0-404.0 kDa	4	0	0		routine	USP34, USP34, ZFHX3, USP34
404.0-406.0 kDa	3	0	0	Actin family	routine	XIRP2 [Actin family], CSMD3, ASPM
408.0-410.0 kDa	2	0	0		routine	VPS13C, SMG1
410.0-412.0 kDa	1	0	0		routine	SMG1
414.0-416.0 kDa	1	0	0		routine	BSN

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
416.0-418.0 kDa	3	0	0		routine	VPS13B, HECTD4, ANK2
418.0-420.0 kDa	1	0	0		routine	VPS13C
420.0-422.0 kDa	1	0	0		routine	DMD
422.0-424.0 kDa	2	0	0		routine	VPS13C, LYST
424.0-426.0 kDa	6	0	0		routine	ADGRV1, DMD, ANK2, LYST, ANK2, DMD
426.0-428.0 kDa	3	0	0	Histone family	routine	ANK2, KMT2A [Histone family], VPS13B
428.0-430.0 kDa	1	0	0		routine	LYST
430.0-432.0 kDa	3	0	0	Histone family	routine	ANK2, KMT2A [Histone family], LYST
432.0-434.0 kDa	2	0	0		routine	ANK2, ANK2
434.0-436.0 kDa	3	0	0	Histone family	routine	LYST, KMT2A [Histone family], TRRAP
436.0-438.0 kDa	4	0	0		routine	TRRAP, ANK2, VPS13B, PKHD1
438.0-440.0 kDa	3	0	0		routine	PKHD1, TRRAP, PKHD1
440.0-442.0 kDa	3	0	0		routine	CMYA5, CMYA5, PKHD1
442.0-444.0 kDa	3	0	0		routine	FRAS1, AKAP9, MUC5B
446.0-448.0 kDa	2	0	0		routine	TNXB, AKAP9
448.0-450.0 kDa	1	0	0		routine	AKAP9
450.0-452.0 kDa	1	0	0	Actin family	routine	PKD1 [Actin family]
452.0-454.0 kDa	2	0	0		routine	DNAH7, DNAH12
456.0-458.0 kDa	1	0	0	Actin family	routine	PKD1 [Actin family]
458.0-460.0 kDa	4	0	1	Heat shock proteins	routine	PRKDC, ANK2, HSPG2 [Heat shock proteins], HSPG2 [Heat shock proteins]
460.0-462.0 kDa	4	0	0	Actin family	routine	PKD1 [Actin family], VPS13D, DNAH7, MUC19
462.0-464.0 kDa	1	0	1		routine	PRKDC
464.0-466.0 kDa	1	0	1		routine	PRKDC
466.0-468.0 kDa	4	0	0	Heat shock proteins	routine	AKAP9, DNAH3, PKHD1L1, HSPG2 [Heat shock proteins]
468.0-470.0 kDa	1	0	0	Heat shock proteins	routine	HSPG2 [Heat shock proteins]
470.0-472.0 kDa	1	0	0		routine	DNAH10
472.0-474.0 kDa	1	0	0		routine	TNXB
474.0-476.0 kDa	2	0	0		routine	UNC13B, DNAH6
476.0-478.0 kDa	1	0	0		routine	STARD9
478.0-480.0 kDa	2	0	0		routine	ANK3, FAT2
480.0-482.0 kDa	2	0	0		routine	HUWE1, VPS13D
482.0-484.0 kDa	6	0	0		routine	HUWE1, ALMS1, STARD9, MUC5B, FAT1, VPS13D
484.0-486.0 kDa	2	0	0		routine	BIRC6, HECTD4
486.0-488.0 kDa	3	0	0		routine	DYNC2H1, HUWE1, DNAH1
488.0-490.0 kDa	1	0	0		routine	VPS13D
490.0-492.0 kDa	2	0	0		routine	VPS13D, VPS13D
492.0-494.0 kDa	2	0	0		routine	DYNC2H1, DYNC2H1
498.0-500.0 kDa	2	0	0		routine	MYCBP2, FAT3
500.0-502.0 kDa	1	0	0		routine	DNAH17
502.0-504.0 kDa	1	0	0		routine	FAT3
504.0-506.0 kDa	2	0	0		routine	LRP1, FAT1
506.0-508.0 kDa	1	0	0		routine	FAT1
508.0-510.0 kDa	3	0	0		routine	DNAH2, MYCBP2, DNAH17
510.0-512.0 kDa	2	0	0		routine	DNAH9, PLEC
512.0-514.0 kDa	1	0	0		routine	SACS
514.0-516.0 kDa	5	0	0		routine	APOB, , DNAH8, APOB, PLEC
516.0-518.0 kDa	4	0	0		routine	DNAH14, MYCBP2, PLEC, SACS
518.0-520.0 kDa	8	0	0	Histone family	routine	SACS, PLEC, PLEC, BIRC6, KMT2C [Histone family], BIRC6, PLEC, LRP2
520.0-522.0 kDa	3	0	0	Histone family	routine	KMT2C [Histone family], MYCBP2, PLEC
522.0-524.0 kDa	2	0	0		routine	DNAH10, MYCBP2
526.0-528.0 kDa	3	0	0		routine	DYNC1H1, HERC2, HERC2
528.0-530.0 kDa	2	0	0		routine	BIRC6, DNAH10
530.0-532.0 kDa	2	0	0		routine	DNAH14, PLEC
532.0-534.0 kDa	3	0	0		routine	HERC1, BIRC6, DYNC1H1
534.0-536.0 kDa	5	0	0	Histone family	routine	PLEC, DYNC1H1, DNAH5, KMT2C [Histone family], PLEC
536.0-538.0 kDa	2	0	0		routine	DNAH11, PCLO
540.0-542.0 kDa	4	0	0	Histone family	routine	KMT2C [Histone family], KMT2C [Histone family], KMT2C [Histone family], KMT2C [Histone family]
542.0-544.0 kDa	5	0	0	Histone family	routine	KMT2C [Histone family], FAT4, BLTP1, SSPO, SSPO
544.0-546.0 kDa	1	0	0		routine	HMCN2
550.0-552.0 kDa	5	0	0		routine	UBR4, ABCA13, RYR3, RYR3, BLTP1
552.0-554.0 kDa	1	0	0		routine	RYR3
554.0-556.0 kDa	1	0	0		routine	BLTP1
556.0-558.0 kDa	1	0	0		routine	ABCA13
558.0-560.0 kDa	1	0	0		routine	BLTP1
560.0-562.0 kDa	1	0	0		routine	PCLO
564.0-566.0 kDa	4	0	0		routine	BLTP1, RYR2, RYR1, RYR1
570.0-572.0 kDa	2	0	0		routine	LOC512869, LOC512869
574.0-576.0 kDa	3	0	0		routine	USH2A, UBR4, UBR4
576.0-578.0 kDa	1	0	0		routine	USH2A
578.0-580.0 kDa	1	0	0		routine	UBR4
588.0-590.0 kDa	1	0	0		routine	LOC512869
592.0-594.0 kDa	1	0	0		routine	AHNAK2
594.0-596.0 kDa	1	0	0		routine	LOC512869
598.0-600.0 kDa	1	0	0		routine	HMCN1
600.0-602.0 kDa	1	0	0		routine	AHNAK
602.0-604.0 kDa	2	0	0	Histone family	routine	KMT2D [Histone family], KMT2D [Histone family]
612.0-614.0 kDa	1	0	0		routine	HMCN1
618.0-620.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
626.0-628.0 kDa	1	0	0		routine	MDN1
630.0-632.0 kDa	1	0	0		routine	MDN1
636.0-638.0 kDa	1	0	0		routine	MDN1
638.0-640.0 kDa	1	0	0		routine	ADGRV1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
668.0-670.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
688.0-690.0 kDa	1	0	0		routine	ADGRV1
760.0-762.0 kDa	1	0	0	Actin family	routine	FSIP2 [Actin family]
772.0-774.0 kDa	1	0	0		routine	NEB
794.0-796.0 kDa	1	0	0		routine	SYNE2
796.0-798.0 kDa	1	0	0		routine	SYNE2
816.0-818.0 kDa	1	0	0		routine	DST
832.0-834.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
836.0-838.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
838.0-840.0 kDa	3	0	0	Actin family	routine	MACF1 [Actin family], MACF1 [Actin family], MACF1 [Actin family]
840.0-842.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
850.0-852.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
854.0-856.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
858.0-860.0 kDa	1	0	0		routine	NEB
862.0-864.0 kDa	1	0	0		routine	DST
882.0-884.0 kDa	1	0	0		routine	NEB
886.0-888.0 kDa	1	0	0		routine	NEB
890.0-892.0 kDa	1	0	0		routine	NEB

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.97	MGMT	P29177	Methylated-DNA--protein-cysteine methyltransferase (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.08		P15507	Morphine-modulating neuropeptide B	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10		A0AAA9S596	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.14		P69206	Morphogenetic neuropeptide	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.38		A0AAA9S118	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.42	HOOK3	P60316	Protein Hook homolog 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.48		A0AAA9TPS1	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.58		A0AAA9TW91	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.74	NUDT7	A0AAA9SML3	Nudix hydrolase 7	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.84		A0AAA9T289	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.91	MXRA7	A0A3Q1MJM7	Matrix-remodeling-associated protein 7	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.92		P15506	Morphine-modulating neuropeptide A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.93		A0ABI0P1B4	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.95		P22779	Cytochrome P450 2A (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.02		A0AAA9TEG0	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.13		A0A3Q1N275	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.18		A0ABI0NYC6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0AAA9SIV3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0AAA9SLK5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0A3Q1LQI6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.22		A0AAA9TM45	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.23		A0A3Q1MRH6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.24		A0AAA9RUI4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.26		A0AAA9S011	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.30		A0AAA9TFU1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.32		A0AAA9S6L4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45		A0AAA9RTD3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45		A0AAA9U0W5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45	RNF165	A0AAA9RSZ2	Arkadia (RNF111) C-terminal like ring finger ubiquitin ligase 2C	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.49		A0ABI0NPU1	Ribosomal protein L32	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.58		A0AAA9RZ15	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.59		A0AAA9TVY5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.62	ABHD5	A0A3Q1LSE8	Abhydrolase domain containing 5, lysophosphatidic acid acyltransferase	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.63	MXRA7	A0AAA9TQV2	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.63		A0A3Q1M3P4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.74		A0ABI0P503	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.77		A0A3Q1MQQ7	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.78		A0AAA9TXY4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.81		A0ABI0P2W9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.85	TEX43	A0AAA9TMI8	Sperm microtubule inner protein 10	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.87		A0AAA9SLS3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.94	TBCA	A0AAA9TXA3	Tubulin folding cofactor A	2.0-4.0 kDa	Tubulin family	configured audit family match: Tubulin family
2.95		A0AAA9SUM1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.95		A0AAA9SIM7	LASP1 neighbor	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.95		A0AAA9RXK3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.98	BIN3	A0AAA9SQE4	Bridging integrator 3	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.01		A0ABI0NQN8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.02	SDHAF2	A0AAA9SCP1	Succinate dehydrogenase complex assembly factor 2	2.0-4.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
3.06	SCT	P63296	Secretin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.08		A0AAA9TF52	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.13	DEFB	F6RMW4	Defensin, beta	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.16		A0A3Q1LY62	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.23		A0AAA9TYS4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.24	SMIM47	A0AAA9TSQ3	Small integral membrane protein 47	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.31	COG5	P83437	Conserved oligomeric Golgi complex subunit 5 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.34		A0ABI0P650	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.46	ZC3H10	A0AAA9SLL4	60S ribosomal protein L41	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.49	TLR4	A0AAA9SFV3	Toll like receptor 4	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.49		A0AAA9TJP5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.59	ACTR3	A0AAA9TM64	Actin related protein 3	2.0-4.0 kDa	Actin family	configured audit family match: Actin family
3.60	CALCA	P01260	Calcitonin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.63	SYNPR	A0A3Q1MSQ5	Synaptoporin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.65		A0A3Q1MX44	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73		A0AAA9T609	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.74	PFDN1	A0AAA9TM01	Prefoldin subunit 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.76	OTX2	A0AAA9TTP9	Orthodenticle homeobox 2	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.76	HRURF	A0AAA9T445	HR upstream open reading frame	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.77	SLN	A0A3Q1MB76	Sarcophilin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.80	LOC613358	A0AAA9SG70	Metallothionein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.88	STRIT1	A0ABI0P1D4	Small transmembrane regulator of ion transport 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.92	BLACAT1	A0AAA9TV63	BLACAT1 overlapping LEMD1 locus	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.03		A0ABI0NT89	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.04	MNT	A0AAA9TLN0	MAX network transcriptional repressor	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.08		A0ABI0P616	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.11	DEFB12	P46170	Beta-defensin 12	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.15	PRM1	A0A452DJ09	Protamine 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.19	OST4	A0AAA9T743	Oligosaccharyltransferase complex subunit 4, non-catalytic	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.19	UBL5	A0AAA9TSK8	Ubiquitin like 5	4.0-6.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
4.28	DEFB1	P46159	Beta-defensin 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.30	ZNF599	A0AAA9SEK2	Zinc finger protein 599	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.32	KALRN	A0AA9TV86	Kalirin RhoGEF kinase	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.34	SMAGP	A0ABI0P578	Small cell adhesion glycoprotein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.36	DEFB8	P46166	Beta-defensin 8	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.38	PDAP1	A0ABI0NSE9	PDGFA associated protein 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.41	MSMB	F1MDF2	Beta-microseminoprotein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.42	PTMA	A0ABI0P2T5	Prothymosin alpha	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.42	LOC132343858	A0A3Q1MKB7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.45	DEFB13	P46171	Beta-defensin 13	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.46	PDCC5	A0AAA9RX22	Programmed cell death 5	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.46	PEX11A	A0AAA9SH95	Peroxisomal biosis factor 11 alpha	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.47		A0AAA9S0W4	SPTY2D1 opposite strand	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.47	MICOS10	A0A3Q1NE13	MICOS complex subunit MIC10	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.48		A0ABI0P513	Metallothionein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.49		A0AAA9S347	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.52	GMFB	A0AAA9SFZ4	Glia maturation factor beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.53	DYNLRB2	A0AAF6Z4W4	Dynein light chain roadblock-type 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.53	UQCC5	A0ABI0P4G5	Ubiquinol-cytochrome c reductase complex assembly factor 5	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
4.60	NREP	A0ABI0NKL8	Neuronal regeneration-related protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.64		A0A3Q1N0R6	Chromodomain helicase DNA binding protein 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.65	DEFB2	P46160	Beta-defensin 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.68	SNAP23	A0AAA9TYI4	Synaptosome associated protein 23	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.72	GOLT1B	A0AAA9TUG0	Golgi transport 1B	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.74	RGMA	A0AAA9T688	Repulsive guidance molecule BMP co-receptor a	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.79	IQSEC2	A0AAA9TFA1	IQ motif and Sec7 domain ArfGEF 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.80	MED31	A0ABI0P675	Mediator of RNA polymerase II transcription subunit 31	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.80	POLR1D	A0ABI0NNR6	RNA polymerase I and III subunit D	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.81	TMSB10	P21752	Thymosin beta-10	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.83	C5H12orf75	A0AA9TXH7	Chromosome 5 C12orf75 homolog	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.84	SLAMF7	A0AAA9S4L5	SLAM family member 7	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.84	DEFB6	P46164	Beta-defensin 6	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.84	RALYL	A0ABI0NP82	RALY RNA binding protein like	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.90		A0AAA9S7Z0	TRAF2 and NCK interacting kinase	4.0-6.0 kDa	Actin family	configured audit family match: Actin family
4.93	C5H12orf75	A0AA9TF77	Chromosome 5 C12orf75 homolog	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.96	GIP	P09680	Gastric inhibitory polypeptide	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01	ASNSD1	E1BB20	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01		A0ABI0NQN9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02	LSM1	A0AA9TS05	LSM1 homolog, mRNA degradation associated	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02	TMSB4	A0A3Q1LXR1	Thymosin beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02	AK6	A0AAF6ZDP6	Adenylate kinase 6	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.03	GABARAPL2	A0ABI0P287	GABA type A receptor associated protein like 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38 δ western blot	40.0-42.0 kDa	ORI reported p38 δ bands were compiled from a source image representing β -actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38 δ -labeled bands were compiled from a β -actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_Glut4	Nrf2, NF-kB p65, PPAR α/δ , Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	cow_UP000009136.fasta
Marker file	cow_marker_genes.tsv
Signaling-prefix file	cow_signaling_prefixes.tsv
Family-pattern file	cow_family_patterns.tsv
Species/report label	Bos taurus